A.J. Hildebrand

Department of Mathematics
University of Illinois at Urbana-Champaign
1409 W. Green Street
Urbana, Illinois 61801-2975
USA
Phone: (217) 244-7721
Fax: (217) 333-9576
Email: ajh@illinois.edu

Office: 13 Illini Hall

Undergraduate Research

I have been involved in guiding undergraduate research since the early 2000s. I have mentored student research under the McNair Scholars Program, directed several NSF Summer REU Programs at Illinois, and currently serve as Director of the NU-UI Mathematics Summer REU Program.

In addition, for the past few years I have been running three or four semester-long Math 492 group research projects each semester, on topics ranging from calculus and geometry to number theory, probability, statistics, and data analytics. Students in my group have presented their work at local and national conferences, participated in outreach activities, won awards for presentations and posters, and have been featured in departmental and university publications; see my Undergraduate Research Page for more.

Math Contests

I have been involved in the UI Math Contest Program since the mid 1980s, and I have been in charge of this program for the past fifteen years. Our program is among the most comprehensive programs of this kind, and it may be the largest such program offered at a public university. Activities include weekly training sessions and practice contests; several local math contests with substantial prize purses; and participation in prestigious national and international competitions. For more information check out the UI Math Contest Website.

Current and Recent Courses

- Math 492, Undergraduate Research in Mathematics (ongoing)

- Math 390, Individual Study. (ongoing)
- Math 199 CHP: Probability and the Real World (Fall 2017). This is a course in the Campus Honors Program, and restricted to Chancellor's Scholars; for more information visit the above link.
- Math 199 CHP: Probability and the Real World. (Spring 2016)
- Math 199 CHP: Probability and the Real World. (Spring 2015)
- Math 347, Section C1, Fundamental Mathematics (Summer 2014)
- Math 241, Section F1H, Calculus III (Honors Section) (Spring 2014)

Course Archive

- Math 213, Section F1, Basic Discrete Mathematics
- Math 370 X, Actuarial Problem Solving
- Math 408, Introduction to Actuarial Statistics I
- Math 453, Section X13/14, Elementary Number Theory
- Math 461, Introduction to Probability
- Math 531, Introduction to Analytic Number Theory
- Math 595, Asymptotic Methods in Analysis
- Math 595, Mathematical Writing

Lecture Notes

- Elementary Number Theory. Math 453 Lecture Notes
- Introduction to Analytic Number Theory. Math 531 Lecture Notes

TeX/LaTeX

- Getting Started with LaTeX
- LaTeX Tips
- TeX Resources

Links

- NU-UI Mathematics Summer REU Program, Summer 2017.
- NU-UI Mathematics Summer REU Program, Summer 2016.
- Number Theory at Illinois: A Conference in Memory of the Batemans and Heini Halberstam. June 5 - 7, 2014.
- Midwest Number Theory Conference for Graduate Students and Recent PhDs, June 3 - 4, 2014.

- Illinois Number Theory Conference, May 27-28, 2011
- Illinois Number Theory Conference, May 21-22, 2010
- REU Program in Experimental Number Theory, UIUC, Summer 2009.
- AMS Sectional Meeting, Urbana, March 27-29, 2009
- Illinois Number Theory Celebration, March 25-26, 2009
- Illinois Number Theory Fest, May 16-20, 2007
- REU Program in Number Theory, UIUC, Summer 2002.
- Illinois Number Theory Conference, May 17-18, 2002
- REU Program in Number Theory, UIUC, Summer 2001.
- Millennial Conference on Number Theory, May 20-26, 2000

Last update: Wed 06 Sep 2017 07:32:07 PM CDT

Math 595AMA: Asymptotic Methods in Analysis Course Web Page

Course Announcement

Lecture Notes

I plan to post notes here once they are in reasonably final form.

- 1. Introduction: What is asymptotic analysis?
- 2. Asymptotic Notations

Last modified Wed 02 Sep 2009 12:49:41 PM CDT A.J. Hildebrand

Math 453: Elementary Number Theory
Definitions and Theorems

(Class Notes, Spring 2011 – A.J. Hildebrand)

Version 5-4-2011

Contents

About these notes **3**

1 Divisibility and Factorization **4**

1.1 Divisibility 4

1.2 Primes 4

1.3 The greatest common divisor 5

1.4 The least common multiple 6

1.5 The Fundamental Theorem of Arithmetic 6

1.6 Primes in arithmetic progressions 8

2 Congruences **9**

2.1 Definitions and basic properties; applications 9

2.2 Linear congruences in one variable 10

2.3 The Chinese Remainder Theorem 10

2.4 Wilson's Theorem 11

2.5 Fermat's Theorem 11

2.6 Euler's Theorem 12

3 Arithmetic functions **13**

3.1 Some notational conventions 13

3.2 Multiplicative arithmetic functions 13

3.3 The Euler phi function and the Carmichael Conjecture 14

3.4 The number-of-divisors functions 15

3.5 The sum-of-divisors functions and perfect numbers 15

3.6 The Moebius function and the Moebius inversion formula 16

3.7 Algebraic theory of arithmetic function 17

3.8 Arithmetic Functions: Summary Table 18

4 Quadratic residues **19**

4.1 Quadratic residues and nonresidues 19

4.2 The Legendre symbol 19

4.3 The law of quadratic reciprocity 20

5 Primitive roots **21**

5.1 The order of an integer 21

5.2 Primitive roots 21

5.3 The Primitive Root Theorem 22

6 Continued fractions **23**

6.1 Definitions and notations 23

6.2 Convergence of infinite continued fractions 23

6.3 Properties of Convergents 24

6.4 Expansions of real numbers into continued fractions 25

7 Topics in Computational Number Theory **26**

7.1 Some Basic Concepts 26

7.2 Primality Tests 26

7.3 The RSA Encryption Scheme 28

About these notes

One purpose of these notes is to serve as a handy reference for homework problems, and especiallly for proof problems. The definitions given here (e.g., of divisibility) are the "authoritative" definitions, and you should use those definitions in proofs. The results stated here are those you are free to use and refer to in proofs; in general, anything else (e.g., a theorem you might have learned in high school) is not allowed.

Another purpose is to serve as a cheat/review sheet when preparing for exams. The definitions and theorems contained in these notes are those you need to know in exams.

Finally, the notes may be useful as a quick reference or refresher on elementary number theory for those taking more advanced number theory classes (e.g., analytic or algebraic number theory).

The notes are loosely based on the Strayer text, though the material covered is pretty standard and can be found, in minor variations, in most undergraduate level number theory texts. The chapters correspond to those in Strayer, but I have made a few small changes in the subdvision of the chapters.

The definitions and results can all be found (in some form) in Strayer, but the numbering is different, and I have made some small rearrangements, for example, combining several lemmas into one proposition, demoting a "theorem" in Strayer to a "proposition", etc. The goal in doing this was to streamline the presentation by having several layers of results, with a clear delineation between the various types of results:

- **Theorems:** Those are the key results, usually with descriptive names attached (e.g., "Fundamental Theorem of Arithmetic"). These results typically have more difficult proofs, often well above homework level.
- **"Starred" theorems:** Results whose statement you should know, but whose proof is beyond the scope of an undergraduate number theory course, are indicated by an asterisk. A typical example is the Prime Number Theorem.
- **Propositions:** A proposition typically collects some simple, but very useful, properties of a concept. The proofs are generally on the easy side, and many (but not all) are at a level that would be reasonable to ask for in an exam.
- **Corollaries:** A corollary is attached to a particular theorem (or proposition), and presents a simple consequence of the theorem, or restates the theorem in a special case. The derivation of a corollary from the corresponding theorem is usually easy, and often immediate.
- **Lemmas:** A lemma is an auxiliary result that is needed (usually) for the proof of a theorem. Lemmas are rarely of interest in their own right, and therefore in general not worth memorizing (in contrast to the other theorem-like structures). Since I do not include the proofs here, I have generally avoided stating lemmas that arise in those proofs. If a result that is stated as "Lemma" in Strayer is important in its own right and worthy of memorizing, I have elevated it to the status of a "Proposition" or "Theorem".

1 Divisibility and Factorization

1.1 Divisibility

Definition 1.1 (Divisibility)**.** *Let $a, b \in \mathbf{Z}$. We say that a* ***divides*** *b (equivalently, a* ***is a divisor of*** *b, or b* ***is divisible by*** *a, or a* ***is a factor of*** *b) if there exists $c \in \mathbf{Z}$ such that $b = ac$. We write $a \mid b$ if a divides b, and $a \nmid b$ if a does* not *divide b.*

Proposition 1.2 (Elementary properties of divisibility)**.**

(i) *(Transitivity) Let $a, b, c \in \mathbf{Z}$. If $a \mid b$ and $b \mid c$, then $a \mid c$.*

(ii) *(Linear combinations) Let $a, b, c \in \mathbf{Z}$. If $a \mid b$ and $a \mid c$, then $a \mid bn + cm$ for any $n, m \in \mathbf{Z}$. In particular, if $a \mid b$ and $a \mid c$, then $a \mid b + c$ and $a \mid b - c$.*

(iii) *(Size of divisors) Let $a, b \in \mathbf{Z}$, with $b \neq 0$. If $a \mid b$, then $|a| \leq |b|$. In particular, any* positive *divisor a of a* positive *integer b must fall in the interval $1 \leq a \leq b$.*

(iv) *(Divisibility and ratios) Let $a, b \in \mathbf{Z}$ with $a \neq 0$. Then $a \mid b$ holds if and only if $\frac{b}{a} \in \mathbf{Z}$.*

Definition 1.3 (Greatest integer function)**.** *For any $x \in \mathbf{R}$, the* ***greatest integer function*** *$[x]$ is defined as the greatest integer m satisfying $m \leq x$. An alternative notation for $[x]$ is $\lfloor x \rfloor$, the* ***floor function****.*

Theorem 1.4 (Division Algorithm)**.** *Given $a, b \in \mathbf{Z}$ with $b > 0$ there exist unique $q, r \in \mathbf{Z}$ such that $a = qb + r$ and $0 \leq r < b$. Moreover, q and r are given by the formulas $q = [a/b]$ and $r = a - [a/b]b$.*

1.2 Primes

Definition 1.5 (Primes and composite numbers)**.** *Let $n \in \mathbf{N}$ with $n > 1$. Then n is called a* ***prime*** *if its only* positive *divisors are* 1 *and n; it is called* ***composite*** *otherwise. Equivalently, n is composite if it can be written in the form $n = ab$ with $a, b \in \mathbf{Z}$ and $1 < a < n$ (and hence also $1 < b < n$); and n is prime otherwise.*

Remark. The number 1 is not classified in this manner, i.e., 1 is neither prime nor composite.

Proposition 1.6 (Existence of prime factors)**.** *Let $n \in \mathbf{N}$ with $n > 1$. Then n has at least one prime factor (possibly n itself); i.e., there exists a prime p with $p \mid n$.*

Proposition 1.7 (Primality test)**.** *Let $n \in \mathbf{N}$ with $n > 1$. Then n is prime if and only if n is not divisible by any prime p with $p \leq \sqrt{n}$.*

Theorem 1.8 (Euclid's Theorem)**.** *There are infinitely many primes.*

Theorem 1.9 (Gaps between primes)**.** *There are arbitrarily large gaps between primes; i.e., for every $n \in \mathbf{N}$, there exist at least n consecutive composite numbers.*

Definition 1.10 (Prime counting function)**.** *Let $x \in \mathbf{R}$ with $x > 0$. Then $\pi(x)$ is the number of primes p with $p \leq x$.*

***Theorem 1.11** (Prime Number Theorem)**.** *The prime counting function $\pi(x)$ satisfies*

$$\lim_{x \to \infty} \frac{\pi(x)}{x/\ln x} = 1.$$

Definition 1.12 (Mersenne and Fermat primes)**.**

(i) *The numbers of the form $M_p = 2^p - 1$, where p is prime, are called* ***Mersenne numbers****; a Mersenne number that is prime is called a* ***Mersenne prime****.*

(ii) *The numbers of the form $F_n = 2^{2^n} + 1$, where $n = 0, 1, \dots$, are called* ***Fermat numbers****; a Fermat number that is prime is called a* ***Fermat prime****.*

Conjectures (Famous conjectures about primes)**.**

(i) ***Mersenne primes:*** *There are infinitely many Mersenne primes.*

(ii) ***Fermat primes:*** *There are only finitely many Fermat primes.*

(iii) ***Twin Prime Conjecture:*** *There infinitely many primes p such that $p+2$ is also prime.*

(iv) ***Goldbach Conjecture:*** *Every* even *integer $n \geq 4$ can be expressed as a sum of two primes (not necessarily distinct), i.e., n can be written in the form $n = p_1 + p_2$, where p_1 and p_2 are primes.*

1.3 The greatest common divisor

Definition 1.13 (Greatest common divisor)**.** *Let $a, b \in \mathbf{Z}$, with a and b not both 0. The* ***greatest common divisor (gcd)*** *of a and b, denoted by $\gcd(a, b)$, or simply (a, b), is defined as the largest among the commons divisor of a and b; i.e.,*

$$(a, b) = \gcd(a, b) = \max\{d : d \mid a \text{ and } d \mid b\}.$$

If $(a, b) = 1$, then a and b are called ***relatively prime*** *or* ***coprime****.*

More generally, the greatest common divisor of n integers $a_1, \dots, a_n$, not all 0, is defined as

$$(a_1, \dots, a_n) = \max\{d : d \mid a_i \text{ for } i = 1, 2, \dots, n\ \}.$$

Proposition 1.14 (Elementary properties of the gcd)**.** *Let $a, b \in \mathbf{Z}$, with a and b not both 0.*

(i) $(a, b) = (-a, b) = (a, -b) = (-a, -b)$.

(ii) $(a, b) = (a + bn, b) = (a, b + am)$ *for any* $n, m \in \mathbf{Z}$.

(iii) $(ma, mb) = m(a, b)$ *for any* $m \in \mathbf{N}$.

(iv) *If $d = (a, b)$, then $(a/d, b/d) = 1$.*

(v) *Let $d \in \mathbf{N}$. Then $d \mid (a, b)$ holds if and only if $d \mid a$ and $d \mid b$.*

Theorem 1.15 (Linear combinations and the gcd)**.** *Let $a, b \in \mathbf{Z}$ with a and b not both 0, and let $d = (a, b)$. Then there exist $n, m \in \mathbf{Z}$ such that $d = na + mb$, i.e., d is a linear combination of a and b with integer coefficients. Moreover, the set of all such linear combinations is exactly equal to the set of integer multiples of d, and d is the least positive element of this set; i.e.,*

$$\{an + bm : n, m \in \mathbf{Z}\} = \{dq : q \in \mathbf{Z}\}$$

and

$$d = \min\{an + bm : n, m \in \mathbf{Z},\ an + bm > 0\}.$$

Theorem 1.16 (Euclidean Algorithm)**.** *Let $a, b \in \mathbf{Z}$ with $a \geq b > 0$. Set $r_0 = a$, $r_1 = b$ and define $r_2, r_3, \dots, r_j$ by iteratively applying the division algorithm as follows, until a remainder 0 is obtained:*

$$\begin{aligned} r_0 &= r_1 q_1 + r_2, \quad 0 < r_2 < r_1 \\ r_1 &= r_2 q_2 + r_3, \quad 0 < r_3 < r_2 \\ &\dots \\ r_{j-2} &= r_{j-1} q_{j-1} + r_j, \quad 0 < r_j < r_{j-1} \\ r_{j-1} &= r_j q_j. \end{aligned}$$

Then (a, b) is equal to the last non-zero remainder, i.e., $(a, b) = r_j$. Moreover, by tracing back the above chain of equations, one obtains an explicit *representation of (a, b) as a linear combination of a and b.*

1.4 The least common multiple

Definition 1.17 (Least common multiple)**.** *Let $a, b \in \mathbf{Z}$, with a and b both nonzero. The* ***least common multiple (lcm) of a and b****, denoted by $[a, b]$, is defined as the smallest positive integer that is divisible by both a and b; i.e.,*

$$[a, b] = \min\{m \in \mathbf{N} : a \mid m \text{ and } b \mid m\}.$$

More generally, the least common multiple of n nonzero integers $a_1, \dots, a_n$ is defined as

$$[a_1, \dots, a_n] = \min\{m \in \mathbf{N} : a_i \mid m \text{ for } i = 1, 2, \dots, n\ \}.$$

Proposition 1.18 (Elementary properties of the lcm)**.** *Let a, b be nonzero integers.*

(i) $[a, b] = [-a, b] = [a, -b] = [-a, -b]$.

(ii) $[ma, mb] = m[a, b]$ *for any* $m \in \mathbf{N}$.

(iii) $[a, b] = \dfrac{|ab|}{(a, b)}$.

(iv) *Let $m \in \mathbf{N}$. Then $[a, b] \mid m$ holds if and only if $a \mid m$ and $b \mid m$.*

1.5 The Fundamental Theorem of Arithmetic

Lemma 1.19 (Euclid's Lemma)**.** *If $a, b \in \mathbf{Z}$, and p is a prime such that $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally, if $a_1, \dots, a_n \in \mathbf{Z}$ and p is a prime such that $p \mid a_1 \cdots a_n$, then there exists an i with $1 \leq i \leq n$ such that $p \mid a_i$.*

Theorem 1.20 (Fundamental Theorem of Arithmetic)**.** *Every integer greater than 1 has a unique factorization into primes; that is, every integer $n > 1$ can be represented in the form*

$$n = \prod_{i=1}^{r} p_i^{\alpha_i},$$

where the p_i are distinct primes, and the exponents α_i are positive integers. Moreover, this representation is unique except for the ordering of the primes p_i.

Notation. *Given an integer $n > 1$, its prime factorization can be represented in any one of the following forms:*

(i) $n = \prod_{i=1}^{s} p_i$, $p_1, \dots, p_s$ *primes (*not necessarily distinct*);*

(ii) $n = \prod_{i=1}^{r} p_i^{\alpha_i}$, $p_1, \dots, p_r$ distinct *primes*, $\alpha_1, \dots, \alpha_r$ positive *integers;*

(iii) $n = \prod_{i=1}^{t} p_i^{\alpha_i}$, $p_1, \dots, p_t$ distinct *primes*, $\alpha_1, \dots, \alpha_t$ nonnegative *integers;*

(iv) $n = \prod_{p \text{ prime}} p^{\alpha_p}$, α_p nonnegative *integers,* $\alpha_p = 0$ *for all but finitely many* p.

In the last form, p runs through all primes, so the product is formally an infinite product. However, since $\alpha_p = 0$ for all but finitely p, all but finitely many terms of the product are 1*, so the product is de facto a finite product.*

The forms (iii) and (iv) are particularly useful when considering the prime factorizations of several integers, since they allow one to express all factorizations with respect to a common "basis" of primes p_i (e.g., the set of all primes that divide at least one *of the given integers, or the set of* all *primes). As an illustration, here are some representations of the prime factorization of $n = 20$:*

$$\begin{aligned} 20 &= 2 \cdot 2 \cdot 5, \\ 20 &= 2^2 \cdot 5^1, \\ 20 &= 2^2 \cdot 3^0 \cdot 5^1 \\ 20 &= 2^2 \cdot 3^0 \cdot 5^1 \cdot 7^0 \cdot 11^0 \cdot 13^0 \cdots \end{aligned}$$

An additional advantage of the forms (iii) and (iv) is that they allow one to represent the integer 1 *(to which the Fundamental Theorem of Arithmetic does not apply)* formally *in the same form, as a product of prime powers, by taking all exponents to be* 0*:*

$$1 = \prod_{i=1}^{t} p_i^0 \quad \textit{or} \quad 1 = \prod_{p} p^0.$$

Proposition 1.21 (Divisibility, gcd, and lcm in terms of prime factorizations)**.** *Let $a, b \in \mathbf{N}$ with prime factorizations (of the form (iii) above) given by*

$$a = \prod_{i=1}^{r} p_i^{\alpha_i}, \quad b = \prod_{i=1}^{r} p_i^{\beta_i},$$

where the p_i are distinct *primes and the exponents α_i and β_i are* nonnegative *integers.*

(i) *Then "a divides b" holds if and only if $\alpha_i \le \beta_i$ for all i.*

(ii) *The gcd and lcm of a and b are given by*

$$(a, b) = \prod_{i=1}^{r} p_i^{\min(\alpha_i, \beta_i)}, \quad [a, b] = \prod_{i=1}^{r} p_i^{\max(\alpha_i, \beta_i)}.$$

1.6 Primes in arithmetic progressions

Definition 1.22 (Arithmetic progression)**.** *A sequence of the form*

$$a, a+b, a+2b, a+3b, \dots, \tag{1.1}$$

where a and b are integers, is called an ***arithmetic progression****.*

***Theorem 1.23** (Dirichlet's Theorem on Primes in Arithmetic Progressions)**.** *Let $a, b \in \mathbf{N}$ with $(a, b) = 1$. Then the arithmetic progression* (1.1) *contains infinitely many primes.*

2 Congruences

2.1 Definitions and basic properties; applications

Definition 2.1 (Congruences)**.** *Let $a, b \in \mathbf{Z}$ and $m \in \mathbf{N}$. We say that a **is congruent to** b **modulo** m, and write $a \equiv b \bmod m$, if $m \mid a-b$ (or, equivalently, if $a = b+mx$ for some $x \in \mathbf{Z}$). The integer m is called the **modulus** of the congruence.*

Proposition 2.2 (Elementary properties of congruences)**.** *Let $a, b, c, d \in \mathbf{Z}$, $m \in \mathbf{N}$.*

(i) *If $a \equiv b \bmod m$ and $c \equiv d \bmod m$, then $a + c \equiv b + d \bmod m$.*

(ii) *If $a \equiv b \bmod m$ and $c \equiv d \bmod m$, then $ac \equiv bd \bmod m$.*

(iii) *If $a \equiv b \bmod m$, then $a^n \equiv b^n \bmod m$ for any $n \in \mathbf{N}$.*

(iv) *If $a \equiv b \bmod m$, then $f(a) \equiv f(b) \bmod m$ for any polynomial $f(n)$ with integer coefficients.*

(v) *If $a \equiv b \bmod m$, then $a \equiv b \bmod d$ for any positive divisor d of m.*

Proposition 2.3 (Congruences as equivalence relation)**.** *Let $m \in \mathbf{N}$. The congruence relation modulo m is an equivalence relation, i.e., satisfies the following properties, for any $a, b, c \in \mathbf{Z}$:*

(i) *(Reflexivity) $a \equiv a \bmod m$.*

(ii) *(Symmetry) If $a \equiv b \bmod m$, then $b \equiv a \bmod m$.*

(iii) *(Transitivity) If $a \equiv b$ and $b \equiv c$, then $a \equiv c$.*

Definition 2.4 (Residue classes)**.** *Let $m \in \mathbf{N}$. The equivalence classes defined by the congruence relation modulo m are called the **residue classes modulo** m. For any $a \in \mathbf{Z}$, $[a]$ denotes the equivalence class to which a belongs, i.e.,*

$$[a] = \{n \in \mathbf{Z} : n \equiv a \bmod m\}.$$

Definition 2.5 (Complete residue system)**.** *A set of integers $r_1, \dots, r_m$ is called a **complete residue system modulo** m, if it contains exactly one integer from each equivalence class modulo m.*

Definition 2.6 (Least nonnegative residue)**.** *Let $m \in \mathbf{N}$. Given any integer n, the **least nonnegative residue of** n **modulo** m is the unique integer r such that $n \equiv r \bmod m$ and $0 \le r < m$; i.e., r is the remainder upon division of n by m by the division algorithm.*

2.2 Linear congruences in one variable

Theorem 2.7 (Solutions of linear congruences in one variable)**.** *Let $a, b \in \mathbf{Z}$ and $m \in \mathbf{N}$, and consider the congruence*

$$ax \equiv b \bmod m. \tag{2.1}$$

Let $d = (a, m)$.

(i) *(Existence of a solution) The congruence (2.1) has a solution $x \in \mathbf{Z}$ if and only if $d \mid b$.*

(ii) *(Number of solutions) Suppose $d \mid b$. Then $ax \equiv b \bmod m$ has exactly d pairwise incongruent solutions x modulo m. The solutions are of the form $x = x_0 + km/d$, $k = 0, 1, \dots, d-1$, where x_0 is a particular solution.*

(iii) *(Construction of a solution) Suppose $d \mid b$. Then a particular solution can be constructed as follows: Apply the Euclidean algorithm to compute $d = (a, m)$, and, working backwards, obtain a representation of d as a linear combination of a and m. Multiply the resulting equation through with (b/d). The new equation can be interpreted as a congruence of the desired type, (2.1), and reading off the coefficient of a gives a particular solution.*

Corollary 2.8. *Let $a \in \mathbf{Z}$ and $m \in \mathbf{N}$. If $(a, m) = 1$, the congruence*

$$ax \equiv 1 \bmod m \tag{2.2}$$

has a unique solution x modulo m; if $(a, m) \neq 1$, the congruence has no solution.

Definition 2.9 (Modular inverses)**.** *A solution x to the congruence (2.2), if it exists, is called a **modular inverse of** a (with respect to the modulus m) and denoted by $\overline{a}$.*

Remark. Note that $\overline{a}$ is not uniquely defined. The definition depends implicitly on the modulus m. In addition, for a given modulus m, $\overline{a}$ is only *unique modulo m*; i.e., any $x \in \mathbf{Z}$ with $x \equiv \overline{a} \bmod m$ is also a a modular inverse of m.

2.3 The Chinese Remainder Theorem

Theorem 2.10 (Chinese Remainder Theorem)**.** *Let $a_1, \dots, a_r \in \mathbf{Z}$ and let $m_1, m_2, \dots, m_r \in \mathbf{N}$ be given such that $(m_i, m_j) = 1$ for $i \neq j$. Then the system*

$$x \equiv a_i \bmod m_i \qquad (i = 1, \dots, r) \tag{2.3}$$

has a unique solution x modulo $m_1 \cdots m_r$.

Corollary 2.11 (Structure of residue systems modulo $m_1 \cdots m_r$)**.** *Let $m_1, \dots, m_r \in \mathbf{N}$ with $(m_i, m_j) = 1$ for $i \neq j$ be given and let $m = m_1 \cdots m_r$. There exists a 1-1 correspondence between complete systems of residues modulo m and r-tupels of complete systems of residues modulo $m_1, \dots, m_r$. More precisely, if, for each i, a_i runs through a complete system of residues modulo m_i, then the corresponding solution x to the simultaneous congruence (2.3) runs through a complete system of residues modulo m.*

2.4 Wilson's Theorem

Theorem 2.12 (Wilson's Theorem)**.** *Let p be a prime number. Then*

$$(p-1)! \equiv -1 \bmod p. \tag{2.4}$$

Theorem 2.13 (Converse to Wilson's Theorem)**.** *If p is an integer ≥ 2 satisfying* (2.4)*, then p is a prime number.*

Remark. The converse to Wilson's Theorem can be stated in contrapositive form as follows: *If n is composite, then $(n-1)!$ is* ***not*** *congruent to -1 modulo n.* In fact, the following much stronger statement holds: *If $n > 4$ and n is composite, then* $(n-1)! \equiv 0 \bmod n$. Thus, for $n > 4$, $(n-1)!$ is congruent to either -1 or 0 modulo n; the first case occurs if and only if n is prime, and the second occurs if and only if n is composite.

2.5 Fermat's Theorem

Theorem 2.14 (Fermat's Little Theorem)**.** *Let p be a prime number. Then, for any integer a satisfying $(a,p) = 1$,*

$$a^{p-1} \equiv 1 \bmod p. \tag{2.5}$$

Corollary 2.15 (Fermat's Little Theorem, Variant)**.** *Let p be a prime number. Then, for any integer a,*

$$a^p \equiv a \bmod p. \tag{2.6}$$

Corollary 2.16 (Inverses via Fermat's Theorem)**.** *Let p be a prime number, and let a be an integer such that $(p,a) = 1$. Then $\overline{a} = a^{p-2}$ is an inverse of a modulo p.*

Remark. In contrast to Wilson's Theorem, Fermat's Theorem does not have a corresponding converse; in fact, there exist numbers p that satisfy the congruence in Fermat's Theorem, but which are compositive. Such "false positives" to the Fermat test are rare, but they do exist, movitating the following definition:

Definition 2.17 (Pseudoprimes and Carmichael numbers)**.** *An integer $p \geq 2$ that is composite, but satisfies the Fermat congruence* (2.5)*, is called a* ***pseudoprime to the base*** *a, or a-**pseudoprime**. A 2-pseudoprime is simply called a **pseudoprime**. An integer p that is a pseudoprime to all bases $a \in \mathbf{N}$ with $(a,p) = 1$ is called a **Carmichael number**.*

2.6 Euler's Theorem

Definition 2.18 (Reduced residue system)**.** *Let $m \in \mathbf{N}$. A set of integers is called a* ***reduced residue system modulo*** *m, if (i) its elements are pairwise incongruent modulo m, and (ii) every integer n with $(n, m) = 1$ is congruent to an element of the set. Equivalently, a reduced residue system modulo m is the subset of a complete residue system consisting of those elements that are relatively prime with m.*

Definition 2.19 (Euler phi-function)**.** *Let $m \in \mathbf{N}$. The* ***Euler phi-function****, denoted by $\varphi(m)$, is defined by*

$$\varphi(m) = \#\{1 \le n \le m : (n, m) = 1\},$$

i.e., $\varphi(m)$ is the number of elements in a reduced system of residues modulo m.

Proposition 2.20. *If $\{r_1, r_2, \dots, r_{\varphi(m)}\}$ is a reduced residue system modulo m, then so is the set $\{ar_1, ar_2, \dots, ar_{\varphi(m)}\}$, for any integer a with $(a, m) = 1$.*

Theorem 2.21 (Euler's generalization of Fermat's theorem)**.** *Let $m \in \mathbf{N}$. Then, for any integer a such that $(a, m) = 1$,*

$$a^{\varphi(m)} \equiv 1 \pmod{m}. \tag{2.7}$$

3 Arithmetic functions

3.1 Some notational conventions

Divisor sums and products: Let $n \in \mathbf{N}$.

- $\sum_{d|n} f(d)$ denotes a sum of $f(d)$, taken over all **positive divisors** d of n.
- $\sum_{p|n} f(p)$ denotes a sum of $f(p)$, taken over all **prime** divisors p of n.
- $\sum_{p^\alpha||n} f(p^\alpha)$ denotes a sum of $f(p^\alpha)$, taken over all **prime powers** p^α that occur in the standard prime factorization of n. (Here the double bar in $p^\alpha||n$ indicates that p^α is the exact power of p dividing n, i.e., $p^\alpha \mid n$, but $p^{\alpha+1} \nmid n$.)
- **Products** over $d \mid n$, $p \mid n$, etc., are defined analogously.

Empty sum/product convention: A sum over an empty set is defined to be 0; a product over an empty set is defined to be 1. Thus, for example, we have

$$\sum_{p^\alpha||1} f(p^\alpha) = 0, \quad \prod_{p^\alpha||1} f(p^\alpha) = 1,$$

since there is no prime power p^α satisfying the condition $p^\alpha||1$.

The above notational conventions greatly simplify the statements of formulas involving arithmetic functions. For example, using these conventions the rather clumsy formula

$$\varphi(n) = \begin{cases} 1 & \text{if } n = 1, \\ \prod_{i=1}^r p_i^{\alpha_i - 1}(p_i - 1) & \text{if } n \ge 2 \text{ and } n = \prod_{i=1}^r p_i^{\alpha_i} \\ & \text{with distinct primes } p_i \text{ and } \alpha_i \in \mathbf{N}, \end{cases}$$

can be rewritten as

$$\varphi(n) = \prod_{p^\alpha||n} p^{\alpha-1}(p-1),$$

without having to introduce subscripts or single out the case $n = 1$. (In the latter case, the product is an empty product, so by the empty product convention, it produces the value 1, which is exactly what we need.)

Sums over 1's ("Bateman summation"): A sum in which each summand is equal to 1 simply counts the number of terms in it; for example, $\sum_{d|n} 1$ is the same as $\#\{d \in \mathbf{N} : d \mid n\}$. While this might seem like a contrived way to represent a counting function, in the context of the general theory of arithmetic functions, such representations are often very useful.

3.2 Multiplicative arithmetic functions

Definition 3.1 (Multiplicative arithmetic function)**.** *A function $f : \mathbf{N} \to \mathbf{C}$ is called an **arithmetic function**. An arithmetic function f is called **multiplicative** if it satisfies the relation*

$$f(n_1 n_2) = f(n_1) f(n_2) \tag{3.1}$$

whenever $((n_1, n_2) = 1)$. *If* (3.1) *holds for* ***all*** $n_1, n_2 \in \mathbf{N}$ *(i.e., without the restriction* $(n_1, n_2) = 1$*), then* f *is called* ***completely multiplicative****.*

Proposition 3.2 (Multiplicative functions and prime factorization)**.** *An arithmetic function* f *that is not identically* 0 *(i.e., such that* $f(n) \neq 0$ *for at least one* $n \in \mathbf{N}$*) is multiplicative if and only if it satisfies*

$$f(n) = \prod_{p^\alpha || n} f(p^\alpha) \quad (n \in \mathbf{N}).$$

In particular, any multiplicative function f *that is not identically* 0 *is uniquely determined by its values* $f(p^\alpha)$ *at prime powers and satisfies* $f(1) = 1$*.*

3.3 The Euler phi function and the Carmichael Conjecture

Definition 3.3 (Euler phi function)**.** *The Euler phi function is defined by*

$$\varphi(n) = \#\{1 \le m \le n : (m, n) = 1\}.$$

Proposition 3.4 (Properties of $\varphi(n)$)**.**

(i) *(Multiplicativity) The Euler phi function is multiplicative (though not completely multiplicative).*

(ii) *(Explicit formula) For any* $n \in \mathbf{N}$*,*

$$\varphi(n) = \prod_{p^\alpha || n} p^{\alpha-1}(p-1) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

(iii) *(Gauss identity)*

$$\sum_{d|n} \varphi(d) = n \quad (n \in \mathbf{N}).$$

Conjecture (Carmichael conjecture)**.** *Given* $n \in \mathbf{N}$*, the equation* $\varphi(x) = n$ *has either no solution* $x \in \mathbf{N}$ *or more than one solution.*

Remark. The Carmichael conjecture has several local (UIUC) connections: Its originator, R.D. Carmichael, spent most of his career as a professor here at the U of I, and the conjecture first appeared as an "exercise" in a textbook on number theory he wrote (and which he presumably assigned to his students). Also, most of the current records on this conjecture are held by Kevin Ford, who earned his PhD here in the mid 1990s and is now back as a professor. In particular, Ford showed that the Carmichael conjecture is true for all $n \le 10^{1000000000}$. Moreover, for any $k \in \mathbf{N}$ *except possibly* $k = 1$, there exist infinitely many $n \in \mathbf{N}$ such that the equation $\varphi(x) = n$ has exactly k solutions $x \in \mathbf{N}$. Thus, only the question of whether multiplicity $k = 1$ can occur remains open, and this is precisely the question addressed by the Carmichael conjecture.

3.4 The number-of-divisors functions

Definition 3.5 (Number-of-divisors function)**.** *The* ***number-of-divisors function*** *is defined by*

$$\nu(n) = \#\{d \in \mathbf{N} : d \mid n\} = \sum_{d|n} 1 \quad (n \in \mathbf{N}).$$

This function is often simply called the ***divisor function****; alternate, and more common, notations for it are $d(n)$ (for "**d**ivisor") and $\tau(n)$ (for "**T**eiler", the German word for "divisor").*

Proposition 3.6 (Properties of $\nu(n)$)**.**

(i) *(Multiplicativity) The function $\nu(n)$ is multiplicative (though not completely multiplicative).*

(ii) *(Explicit formula) For any $n \in \mathbf{N}$,*

$$\nu(n) = \prod_{p^\alpha || n} (\alpha + 1)$$

3.5 The sum-of-divisors functions and perfect numbers

Definition 3.7. *Sum-of-divisors function The* ***sum-of-divisors function*** *is defined by*

$$\sigma(n) = \sum_{d|n} d \quad (n \in \mathbf{N}).$$

Proposition 3.8 (Properties of $\sigma(n)$)**.**

(i) *(Multiplicativity) The function $\sigma(n)$ is multiplicative (though not completely multiplicative).*

(ii) *(Explicit formula) For any $n \in \mathbf{N}$,*

$$\sigma(n) = \prod_{p^\alpha || n} \frac{p^{\alpha+1} - 1}{p - 1}$$

Definition 3.9 (Perfect numbers)**.** *An positive integer n is called* ***perfect*** *if it is equal to the sum of its positive divisors d $|n$, with $1 \le d < n$ (i.e., not counting $d = n$). Equivalently, n is perfect if and only if $\sigma(n) = 2n$.*

Example. The first 4 perfect numbers are $6(= 1+2+3)$, $28(= 1+2+4+7+14)$, 496, and 8128.

Theorem 3.10 (Characterization of even perfect numbers)**.** *An* even *positive integer n is perfect if and only if it is of the form*

$$n = 2^{p-1}(2^p - 1),$$

where $2^p - 1$ is a Mersenne prime.

Corollary 3.11. *There exist infinitely many even perfect numbers if and only if there exist infinitely many Mersenne primes.*

Example. The above four perfect numbers $6, 28, 496, 8128$ correspond to the first four Mersenne primes, $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$, and $2^7 - 1 = 127$.

3.6 The Moebius function and the Moebius inversion formula

Definition 3.12 (Moebius function)**.** *The* ***Moebius function*** *is defined by*

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^r & \text{if } n = p_1 p_2 \dots p_r \text{ with distinct primes } p_i, \\ 0 & \text{if } n \text{ is not squarefree, i.e., divisible by a prime power } p^\alpha \text{ with } \alpha > 1. \end{cases}$$

Remark. A very similar function is the **Liouville function** $\lambda(n)$, which is defined as $(-1)^r$, where r is the *total* number of prime factors of n, with multiple prime factors counted multiple times. If n is squarefree, then $\lambda(n) = \mu(n)$, but if n is not squarefree, then $\mu(n) = 0$, while $\lambda(n) = \pm 1$ depending on whether n has an even or an odd number of prime factors.

While the Liouville function has a simpler definition and may seem the more natural of the two functions, the Moebius function is more useful and more important because of results such those below (which would not be valid for the Liouville function).

Proposition 3.13 (Properties of $\mu(n)$)**.**

(i) *(Multiplicativity) The Moebius function is multiplicative (though not completely multiplicative).*

(ii) *(Moebius function identity)*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 3.14 (Moebius inversion formula)**.** *If f and g are arithmetic functions satisfying*

$$f(n) = \sum_{d|n} g(d) \quad (n \in \mathbf{N}),$$

then

$$g(n) = \sum_{d|n} f(d)\mu(n/d) = \sum_{d|n} \mu(d) f(n/d) \quad (n \in \mathbf{N}).$$

3.7 Algebraic theory of arithmetic function

In this section we develop an algebraic theory of arithmetic functions based on the notion of a **Dirichlet product**. This allows us to restate many of the definitions and results encountered earlier in a simple, elegant, and natural form.

We begin by defining some "trivial" arithmetic functions that are needed in this theory and which can be used to build up other arithmetic functions.

Definition 3.15 (Trivial arithmetic functions)**.** *The* ***unit function*** $\mathbf{1}$*,* ***identity function*** $\mathbf{id}$*, and* ***delta function*** δ *are the arithmetic functions defined as follows:*

$$\begin{aligned} \mathbf{1}(n) &= 1 \quad (n \in \mathbf{N}), \\ \mathbf{id}(n) &= n \quad (n \in \mathbf{N}), \\ \delta(n) &= \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

All three of these functions are multiplicative (in fact, completely multiplicative).

Definition 3.16 (Dirichlet product of arithmetic functions)**.** *Given two arithmetic functions f and g, the* ***Dirichlet product*** $f \star g$ *is the arithmetic function defined by*

$$(f \star g)(n) = \sum_{d|n} f(d)g(n/d) \quad (n \in \mathbf{N}).$$

Proposition 3.17 (Algebraic properties of Dirichlet product)**.** *Let f, g, h be arithmetic functions.*

(i) *(Commutativity) $f \star g = g \star f$.*

(ii) *(Associativity) $(f \star g) \star h = f \star (g \star h)$.*

(iii) *(Identity element) $f \star \delta = \delta \star f = f$, where δ is defined as above, i.e., $\delta(1) = 1$ and $\delta(n) = 0$ if $n > 1$.*

(iv) *(Dirichlet inverse) If $f(1) \neq 0$, then f has a unique Dirichlet inverse f^{*-1}, in the sense that $f \star f^{*-1} = \delta$.*

(v) *(Preservation of multiplicativity) The Dirichlet product of two multiplicative functions is multiplicative.*

(vi) *(Multiplicativity of inverse) The Dirichlet inverse of a multiplicative function is multiplicative.*

Using the Dirichlet product notation, we can now restate many of the definitions, identities, and theorems on arithmetic functions encountered earlier in a simple, elegant, very natural, and easy-to-remember form.

Proposition 3.18 (Drichlet product versions of identities for arithmetic functions)**.**

(i) ***Gauss identity:*** $\varphi \star \mathbf{1} = \mathbf{id}$

(ii) ***Definition of the divisor function:*** $\nu = \mathbf{1} \star \mathbf{1}$

(iii) ***Definition of the sum-of-divisors function:*** $\sigma = \mathbf{1} \star \mathbf{id}$

(iv) ***Moebius function identity:*** $\mu \star \mathbf{1} = \delta$

(v) ***Moebius inversion formula:*** *If $f = g \star \mathbf{1}$, then $g = f \star \mu = \mu \star f$.*

3.8 Arithmetic Functions: Summary Table

Function	**value at** $n(\in \mathbf{N})$	**value at a prime** p	**value at a prime power** p^α	**properties**
$\delta(n)$ (delta function)	1 if $n = 1$, 0 else	0	0	completely multiplicative, $\delta \star f = f \star \delta = f$, identity element for Dirichlet product
$\mathbf{1}(n)$ (unit function)	1	1	1	completely multiplicative
$\mathbf{id}(n)$ (identity function)	n	p	p^α	completely multiplicative
$\mu(n)$ (Moebius function)	1 if $n = 1$, $(-1)^r$ if $n = \prod_{i=1}^r p_i$ (p_i distinct), 0 otherwise	-1	-1 if $\alpha = 1$, 0 if $\alpha > 1$	multiplicative, $\mu \star \mathbf{1} = \delta$ (Dirichlet inverse of $\mathbf{1}$)
$\nu(n)\,(= d(n) = \tau(n))$ (number-of-divisors function)	$\#\{d \in \mathbf{N} : d \mid n\}$	2	$\alpha + 1$	multiplicative, $\nu = 1 \star 1$
$\varphi(n)$ (Euler phi function)	$\#\{1 \le m \le n : (m, n) = 1\}$	$p - 1$	$p^{\alpha-1}(p-1)$	multiplicative, $\varphi \star \mathbf{1} = \mathbf{id}$ (Gauss identity)
$\sigma(n)$ (sum-of-divisors function)	$\sum_{d\mid n} d$	$p + 1$	$\dfrac{p^{\alpha+1} - 1}{p - 1}$	multiplicative, $\sigma = \mathbf{id} \star \mathbf{1}$

Table 1: Summary of important arithmetic functions

4 Quadratic residues

4.1 Quadratic residues and nonresidues

Definition 4.1 (Quadratic residues and nonresidues)**.** *Let $m \in \mathbf{N}$ and $a \in \mathbf{Z}$ be such that $(a, m) = 1$. Then a called a **quadratic residue modulo** m if the congruence*

$$x^2 \equiv a \bmod m \tag{4.1}$$

*has a solution (i.e., if a is a "perfect square modulo m"), and a is called a **quadratic nonresidue modulo** m if* (4.1) *has no solution.*

Remarks. (i) Note that, by definition, integers a that do not satisfy the condition $(a, m) = 1$ are not classified as quadratic residues or nonresidues. In particular, 0 is considered neither a quadratic residue nor a quadratic nonresidue (even though, for $a = 0$, (4.1) has a solution, namely $x = 0$).

(ii) While the definition of quadratic residues and nonresidues allows the modulus m to be an arbitrary positive integer, in the following we will focus exclusively on the case when m is an *odd prime* p.

Proposition 4.2 (Number of solutions to quadratic congruences)**.** *Let p be an odd prime, and let $a \in \mathbf{Z}$ with $(a, p) = 1$.*

(i) *If a is a quadratic nonresidue modulo p, the congruence* (4.1) *has no solution.*

(ii) *If a is a quadratic residue modulo p, the congruence* (4.1) *has exactly two incongruent solutions x modulo p. More precisely, if x_0 is one solution, then a second, incongruent, solution is given by $p - x_0$.*

Proposition 4.3 (Number of quadratic residues and nonresidues)**.** *Let p be an odd prime. Then among the integers $1, 2, \ldots, p-1$, exactly half (i.e., $(p-1)/2$) are quadratic residues modulo p, and exactly half are quadratic nonresidues modulo p.*

4.2 The Legendre symbol

Definition 4.4 (Legendre symbol)**.** *Let p be an odd prime, and let a be an integer with $(a, p) = 1$ (or, equivalently, $p \nmid a$). The **Legendre symbol of** a **modulo** p, denoted by $\left(\frac{a}{p}\right)$, is defined as*

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

Remark. Note that the modulus in this definition, and in all results below, is restricted to odd primes (i.e., a prime other than 2). One can extend the definition, and most of the results, to composite moduli, but things get a lot more complicated then.

Proposition 4.5 (Properties of the Legendre Symbol)**.** *Let p be an odd prime, and let $a, b \in \mathbf{Z}$ with $(a, p) = 1$ and $(b, p) = 1$.*

(i) *(Periodicity in numerator) If $a \equiv b \bmod p$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.*

(ii) *(Complete multiplicativity in numerator)* $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.

(iii) *(Value at squares)* $\left(\dfrac{a^2}{p}\right) = 1$.

(iv) *(Value at -1)* $\left(\dfrac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \bmod 4, \\ -1 & \text{if } p \equiv 3 \bmod 4. \end{cases}$

(v) *(Value at 2)* $\left(\dfrac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv 1, 7 \bmod 8, \\ -1 & \text{if } p \equiv 3, 5 \bmod 8. \end{cases}$

Proposition 4.6 (Euler's Criterion)**.** *Let p be an odd prime, and let $a \in \mathbf{Z}$ with $(a, p) = 1$. Then a is a quadratic residue modulo p if $a^{(p-1)/2} \equiv 1 \bmod p$, and a quadratic nonresidue if $a^{(p-1)/2} \equiv -1 \bmod p$; equivalently,*

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \bmod p.$$

4.3 The law of quadratic reciprocity

Theorem 4.7 (Quadratic reciprocity law (Gauss 1795))**.** *Let p and q be distinct odd primes. Then*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

Equivalently,

$$\left(\frac{p}{q}\right) = \begin{cases} -\left(\dfrac{q}{p}\right) & \text{if } p \equiv 3 \bmod 4 \text{ and } q \equiv 3 \bmod 4, \\ \left(\dfrac{q}{p}\right) & \text{otherwise.} \end{cases}$$

Remarks. (i) The first form of the reciprocity law is the cleaner and more elegant form, and the one in which it the law is usually stated. However, for applications, the second form is more useful. In this form the law says that numerator and denominator in a Legendre symbol (assuming both are distinct odd primes) can be interchanged in all cases except when both numerator and denominator are congruent to 3 modulo 4, in which case the sign of the Legendre symbol flips after interchanging numerator and denominator. Put differently, this form states that p is a quadratic residue modulo q if and only if q is a quadratic residue modulo p, except in the case when both p and q are congruent to 3 modulo 4; in the latter case p is a quadratic residue modulo q if and only if q is a quadratic nonresidue modulo p.

(ii) Note that the reciprocity law requires numerator and denominator to be distinct odd primes. In particular, it does not apply directly to cases where the numerator is composite, negative, or an even number. However, these cases can be reduced to the prime case using the multiplicativity of the Legendre symbol along with the special values at -1 and 2 (see Proposition 4.5):

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \bmod 4, \\ -1 & \text{if } p \equiv 3 \bmod 4. \end{cases}$$

and

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv 1, 7 \bmod 8, \\ -1 & \text{if } p \equiv 3, 5 \bmod 8. \end{cases}$$

In fact, these last two relations are called the **First Supplementary Law** and **Second Supplementary Law**, as they "supplement" the quadratic reciprocity law.

5 Primitive roots

5.1 The order of an integer

Definition 5.1 (Order of an integer)**.** *Let $m \in \mathbf{N}$ and $a \in \mathbf{Z}$ be such that $(a, m) = 1$. The **order of** a **modulo** m, denoted by $\operatorname{ord}_m a$, is the least positive integer k such that*

$$a^k \equiv 1 \bmod m. \tag{5.1}$$

In order for this definition to make sense, there has to be at least one positive integer k for which (5.1) holds. The existence of such a k is guaranteed by Euler's Theorem (Theorem 2.21), which states that, under the same assumptions on m and a as in the definition, (5.1) holds for $k = \varphi(m)$. Thus, the order $\operatorname{ord}_m a$ is well-defined, and it is at most equal to $\varphi(m)$.

Proposition 5.2 (Properties of an order)**.** *Let $m \in \mathbf{N}$ and $a \in \mathbf{Z}$ be such that $(a, m) = 1$, and let $\operatorname{ord}_m a$ be the order of a modulo m. Then:*

(i) *(Periodicity) If $b \equiv a \bmod m$, then $\operatorname{ord}_m b = \operatorname{ord}_m a$.*

(ii) *(Relation to Euler phi) $\operatorname{ord}_m a$ is a divisor of $\varphi(m)$.*

(iii) *(Characterization of "good" exponents) The set of positive integers k for which the congruence (5.1) holds consists exactly of the positive integer multiples of $\operatorname{ord}_m a$.*

(iv) *(Order of powers of a) For any positive integer i,*

$$\operatorname{ord}_m a^i = \frac{\operatorname{ord}_m a}{(\operatorname{ord}_m a, i)}.$$

In particular, $\operatorname{ord}_m a^i = \operatorname{ord}_m a$ if and only if $(\operatorname{ord}_m a, i) = 1$.

Proposition 5.3 (Number of elements of given order)**.** *Let p be an odd prime. Then the possible orders of integers modulo p are exactly the positive divisors of $p - 1 (= \varphi(p))$. Moreover, given any positive divisor $d \mid p - 1$, there exist exactly $\varphi(d)$ incongruent integers a with $\operatorname{ord}_p a = d$.*

5.2 Primitive roots

The question when the order of an integer a modulo m is equal to its maximal possible value, the "Euler order" $\varphi(m)$, motivates the following definition.

Definition 5.4 (Primitive root)**.** *Let $m \in \mathbf{N}$ and $a \in \mathbf{Z}$ be such that $(a, m) = 1$. Then a is called a **primitive root modulo** m if $\operatorname{ord}_m a = \varphi(m)$, i.e., if the order of a is equal to the maximal possible value.*

Proposition 5.5 (Primitive roots and reduced systems of residues)**.** *Let $m \in \mathbf{N}$, and suppose r is a primitive root modulo m. Then the set*

$$\{r, r^2, \dots, r^{\varphi(m)}\}$$

is a system of reduced residues modulo m. That is, the elements in this set are pairwise incongruent modulo m, and every integer a with $(a, m) = 1$ is congruent modulo m to an element in the above set.

5.3 The Primitive Root Theorem

Theorem 5.6 (Existence of Primitive Roots)**.** *Let m be a positive integer. Then there exists a primitive root modulo m if and only if m has one of the following forms:*

(i) $m = p^\alpha$, *where p is an odd prime and $\alpha \in \mathbf{N}$.*

(ii) $m = 2p^\alpha$, *where p is an odd prime and $\alpha \in \mathbf{N}$.*

(iii) $m = 1, 2, 4$.

Theorem 5.7 (Number of primitive roots)**.** *Let m be of one of the forms in the Primitive Root Theorem, so that there exists at least one primitive root modulo m. Then there exist exactly $\varphi(\varphi(m))$ incongruent primitive roots modulo m.*

6 Continued fractions

6.1 Definitions and notations

Definition 6.1 (Continued fractions). *A finite or infinite expression of the form*

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \dots}}, \tag{6.1}$$

*where the a_i are real numbers, with $a_1, a_2, \dots > 0$, is called a **continued fraction** (c.f.). The numbers a_i are called the **partial quotients** of the c.f.*

The continued fraction (6.1) *is called **simple** if the partial quotients a_i are all integers. It is called **finite** if it terminates, i.e., if it is of the form*

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \dots + \cfrac{1}{a_n}}}, \tag{6.2}$$

*and **infinite** otherwise.*

Notation (Bracket notation for continued fractions). *The continued fractions* (6.1) *and* (6.2) *are denoted by $[a_0, a_1, a_2, \dots]$ and $[a_0, a_1, a_2, \dots, a_n]$, respectively. In particular,*

$$[a_0] = a_0, \quad [a_0, a_1] = a_0 + \frac{1}{a_1}, \quad [a_0, a_1, a_2] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2}}, \quad \dots$$

Remarks. (i) Note that the first term, a_0, is allowed to be negative or 0, but all subsequent terms a_i must be positive. This requirement ensures that there are no zero denominators and that any finite c.f. (6.2), and all of its convergents, are well-defined.

(ii) In the sequel we will focus on the case of simple c.f.'s, i.e., c.f.'s where all partial quotients are integers.

6.2 Convergence of infinite continued fractions

Definition 6.2 (Convergents). *The **convergents** of a (finite or infinite) c.f. $[a_0, a_1, a_2, \dots]$ are defined as*

$$C_0 = [a_0],\ C_1 = [a_0, a_1],\ C_2 = [a_0, a_1, a_2], \dots$$

If the c.f. is simple, its convergents C_i represent rational numbers, denoted by

$$C_i = \frac{p_i}{q_i},$$

where p_i/q_i is in reduced form.

Definition 6.3 (Convergence of infinite continued fractions). *An infinite c.f. $[a_0, a_1, a_2, \dots]$ is called **convergent** if its sequence of convergents $C_i = [a_0, a_1, \dots, a_i]$ converges in the usual sense, i.e., if the limit*

$$\alpha = \lim_{i \to \infty} C_i = \lim_{i \to \infty} [a_0, a_1, \dots, a_i]$$

exists (and is a real number). In this case, we say that the continued fraction $[a_0, a_1, a_2, \dots]$ ***represents*** *the number* α*, or is a* ***continued fraction expansion*** *of* α*, and we write*

$$\alpha = [a_0, a_1, a_2, \dots].$$

Theorem 6.4 (Convergence of infinite simple c.f.'s)**.** *Any infinite simple c.f.* $[a_0, a_1, \dots]$ *is convergent and thus represents some real number.*

6.3 Properties of Convergents

Proposition 6.5 (Formulas for p_i and q_i)**.** *Let* $\alpha = [a_0, a_1, \dots]$ *be a simple c.f. with convergents* $C_i = [a_0, a_1, \dots, a_i] = \frac{p_i}{q_i}$.

(i) ***Recursion formula:*** *The numbers* p_i *and* q_i *are given by the recurrence*

$$p_i = a_i p_{i-1} + p_{i-2},$$
$$q_i = a_i q_{i-1} + q_{i-2}$$

for $i = 1, 2, \dots$*, along with the initial conditions* $p_0 = a_0, p_{-1} = 1, q_0 = 1, q_{-1} = 0$.

(ii) ***Matrix representation:*** *For* $i = 0, 1, 2, \dots$

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_i & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_i & p_{i-1} \\ q_i & q_{i-1} \end{pmatrix}$$

Theorem 6.6 (Properties of convergents)**.** *The convergents* $C_i = p_i/q_i$ *of an infinite simple continued fraction* $\alpha = [a_0, a_1, a_2, \dots]$ *satisfy:*

(i) $(p_i, q_i) = 1$ *for* $i = 0, 1, \dots$*; i.e., the fractions* p_i/q_i *are reduced.*

(ii) $q_1 < q_2 < \cdots$*; i.e., for* $i \geq 1$*, the denominators* q_i *are strictly increasing.*

(iii) $C_0 < C_2 < C_4 < \cdots < \alpha < \cdots < C_5 < C_3 < C_1$*. That is, the even-indexed convergents form an increasing sequence, while the odd-indexed convergents form a decreasing sequence, with the value of the c.f. sandwiched between both sequences.*

(iv) $C_{i+1} - C_i = \dfrac{(-1)^i}{q_i q_{i+1}}$ *for* $i = 0, 1, 2, \dots$.

(v) $\left|\dfrac{p_i}{q_i} - \alpha\right| < \dfrac{1}{q_i q_{i+1}}$ *for* $i = 0, 1, 2, \dots$.

(vi) ***Best approximation property:*** *For any rational number* a/b *with* $a \in \mathbf{Z}$*,* $b \in \mathbf{N}$*, and* $1 \leq b \leq q_i$*,*

$$\left|\frac{p_i}{q_i} - \alpha\right| \leq \left|\frac{a}{b} - \alpha\right|,$$

with equality if and only if $a/b = p_i/q_i$*. That is, the convergent* p_i/q_i *is the best-possible approximation to* α *among all rational numbers with the same or smaller denominator.*

6.4 Expansions of real numbers into continued fractions

Proposition 6.7 (Continued fraction algorithm)**.** *Given a real number α, define successively real numbers $\alpha_0, \alpha_1, \dots$, and integers $a_0, a_1, \dots$ by*

$$\begin{aligned} \alpha_0 &= \alpha, & a_0 &= [\alpha_0], \\ \alpha_1 &= \frac{1}{\alpha_0 - [\alpha_0]}, & a_1 &= [\alpha_1], \\ \alpha_2 &= \frac{1}{\alpha_1 - [\alpha_1]}, & a_2 &= [\alpha_2], \\ &\dots & &\dots \end{aligned}$$

where $[x]$ denotes the integer part of x (i.e., the "floor function"). Stop the algorithm if α_n is an integer (and thus $a_n = \alpha_n$); otherwise continue indefinitely. Then $[a_0, a_1, \dots]$ is a simple c.f. that represents the number α. Moreover, for any $i \ge 0$ we have

$$\alpha_i = [a_i, a_{i+1}, \dots], \quad \alpha = [a_0, a_1, \dots a_{i-1}, \alpha_i].$$

Theorem 6.8 (Continued fraction expansion of rational numbers)**.** *Any finite simple c.f. represents a rational number. Conversely, any rational number α can be expressed as a simple finite c.f. $\alpha = [a_0, a_1, \dots, a_n]$. Moreover, under the requirement that $a_n > 1$, this representation is unique. Thus, there is a one-to-one correspondence between rational numbers and finite simple c.f.'s with last partial quotient greater than 1.*

Theorem 6.9 (Continued fraction expansion of irrational numbers)**.** *Any infinite simple c.f. represents an irrational number. Conversely, any irrational number α can be expressed as a simple* infinite *c.f. $\alpha = [a_0, a_1, a_2, \dots]$, and this representation is unique. Thus, there is a one-to-one correspondence between irrational numbers and infinite simple c.f.'s.*

Theorem 6.10 (Continued fraction expansion of quadratic irrationals)**.** *The c.f. expansion of a quadratic irrational (i.e., a solution of a quadratic equation with integer coefficients) is eventually periodic, i.e., of the form*

$$[a_0, \dots, a_N, \overline{a_{N+1}, \dots, a_{N+p}}],$$

where the bar indicates the periodic part. Conversely, any infinite simple c.f. that is eventually periodic represents a quadratic irrational. Thus, there is a one-to-one correspondence between quadratic irrationals and infinite, eventually periodic simple c.f.'s.

7 Topics in Computational Number Theory

7.1 Some Basic Concepts

Running Time: A fundamental notion that allows one to quantify and compare the efficieny of algorithms. The running time of an algorithm is the number of "basic" operations it takes to complete given an input of "size" n.

Depending on the context, "basic" operation can refer to simple arithmetic operations like addition or multiplication, or single bit operations (AND, OR, etc.), or individual CPU instructions. For the applications we consider here, it does not matter much which interpretation one uses.

The "size" of an input is measured in terms of the number of bits, or digits. Thus, for example, an integer with 1000 digits (decimal or binary) would have size around 1000. In general, if N is a large integer, then its size n is proportional to $\log N$. *It is important to keep in mind that the appropriate yardstick when measuring running times is not the size of the integer itself, but its logarithm.*

Polynomial Time: A key benchmark for running times is "polynomial time", i.e., a running time that, for an input of size n, is of order of a fixed power of n. In a sense, this is best-possible, as reading in an input of size n in bit-by-bit fashion already requires n bit operations. Finding algorithms that are polynomial time, or proving that no such algorithms exist for a particular problem, is one of the fundamental problems in the theory of algorithms.

Factoring versus Multiplying: By the Fundamental Theorem of Arithmetic, there is a one-to-one correspondence between finite tuples of prime factors $(p_1, \dots, p_r)$ with $p_1 \le \cdots \le p_r$ and integers $n \ge 2$, given by the bijection $(p_1, \dots, p_r) \leftrightarrow n = p_1 \dots p_r$. However, from a computational point of view the two directions of this bijection are vastly different: Going from left to right simply requires multiplying together the prime factors p_i, a trivial operation that can be carried out in polynomial time. By contrast, the reverse direction requires factoring an integer into its prime factors and is a computationally much harder task. The fact that one direction in this equivalence is easy from a computational point of view, while the other is hard, is the basis of most modern cryptographic schemes.

Primality Testing versus Factoring: Another crucial difference from a computational point of view lies between primality testing algorithms and factoring algorithms. A factoring algorithm takes as input a number n and produces as output its complete prime factorization. By contrast, a primality testing algorithm produces as output PRIME or COMPOSITE (and possibly also INCONCLUSIVE), without exhibiting any factors.

Computationally, primality testing is much easier, and faster, than factoring. There are algorithms known that test primality in polynomial time. By contrast, there are no known polynomial time factoring algorithms, and it is conjectured that none exist.

7.2 Primality Tests

Trial Division: Let $n \ge 2$. For $d = 2, 3, \dots, \lfloor \sqrt{n} \rfloor$, check if $d \mid n$ (e.g., using division with remainder).

- If $d \mid n$ for some $2 \le d \le \sqrt{n}$, then n is composite, and d is a proper divisor of n.
- If $d \nmid n$ for all $2 \le d \le \sqrt{n}$, then n is prime.

Comments: As a general primality test, Trial Division is not practical since it requires about $\sqrt{n}$ operations to determine the primality of an integer. However, it is useful as an initial test to detect integers that have a small prime factor and eliminate these from further consideration

before applying more sophisticated tests. Also, in contrast to all of the tests below, Trial Division produces explicit factors, and it can be used recursively to completely factor a composite number.

Wilson's Test: Let $n \geq 2$. Compute $(n-1)! \bmod n$.

- If $(n-1)! \not\equiv -1 \bmod n$, then n is composite.
- If $(n-1)! \equiv -1 \bmod n$, then n is prime.

Comments: Unlike the Fermat Test below, Wilson's Test is an "if and only if" statement, and thus provides an ironclad guarantee of compositeness or primality. As a practical primality test, however, it is not very useful since there is no efficient (fast) way to compute $(n-1)!$ modulo n.

Fermat Test: Let $n \geq 2$. Pick an integer a with $(a, n) = 1$, and compute $a^{n-1} \bmod n$.

- If $a^{n-1} \not\equiv 1 \bmod n$, then n is composite.
- If $a^{n-1} \equiv 1 \bmod n$, then n is likely (but not guaranteed) prime.

Comments: The Fermat test is very fast: Using repeated squaring to compute the powers a^{n-1} modulo n, it can be performed in polynomial time.

The test has no "false negatives": If a number n fails the test, i.e., if $a^{n-1} \not\equiv 1 \bmod n$, then n is *guaranteed* composite.

The test does have "false positives": There are numbers n that pass the Fermat test, i.e., satisfy $a^{n-1} \equiv 1 \bmod n$, but are composite. Such numbers are called *pseudoprimes* (to base a). Luckily, false positives are extremely rare. For example, out of the first billion integers n that pass the base-2 Fermat test (i.e., satisfy $2^{n-1} \equiv 1 \bmod n$), only about $20,000$ are false positives (i.e., pseudoprimes). Thus, in this range, the test is 99.998% accurate. This makes the Fermat test very useful as a test that identifies "probable primes", i.e., numbers that are, with very high probability, prime, and which can then be subjected to further tests (such as Lucas' Test below).

Lucas Test: Let $n \geq 2$. Let a be an integer with $(a, n) = 1$, and suppose the following two conditions hold:

- $a^{n-1} \equiv 1 \bmod n$.
- $a^{(n-1)/q} \not\equiv 1 \bmod n$ for every prime divisor $q \mid n-1$.

Then n is prime.

Comments: This test fixes the "hole" in Fermat's test: The first condition is simply the Fermat test, while the second, additional, condition guarantees that n is indeed a prime.

The downside of Lucas' test is that, in order to verify that this additional condition is satisfied, one first needs to find all prime factors of $n-1$. For general integers n, this is a much harder problem than the problem of testing n for primality. However, the test can be useful for integers for which the prime factorization of $n-1$ is known in advance, such as Fermat numbers. In the fact, in this special case conditions in Lucas' test can be further simplified to yield the following test for the primality of Fermat numbers:

Pépin's Test: Let $F_m = 2^{2^m} + 1$ be the m-th Fermat number. Then F_m is prime if and only if

$$3^{(F_m-1)/2} \equiv -1 \bmod F_m.$$

Comments: This test, and variations of it (for example, with 3 replaced by 5), is useful in testing Fermat numbers for primality. A key feature of this test is the "if and only if" character: If the stated congruence holds, F_m is guaranteed to be prime; if it does not hold, F_m is guaranteed to be composite.

7.3 The RSA Encryption Scheme

Encryption basics: An encryption scheme takes a message M and creates an encrypted version E of this message by applying an appropriate one-to-one function, f, the *encryption function*, to M: $E = f(M)$. The associated decryption scheme works the same way, with the encrypted message E as input, and the inverse functing, f^{-1}, as a *decryption function* that recovers the original message: $M = f^{-1}(E)$.

A classic example of an encryption scheme is the "Caesar cipher", which shifts each letter in the alphabet forward by 3, so that A gets mapped to D, B maps to E, C maps to F, etc.

Private key encryption versus public key encryption: In the example of the Caesar cipher, anyone who knows the encryption function ("shift the letter forward by 3") is able to deduce the corresponding decryption function ("shift the letter backward by 3") and thus can decrypt any messages encrypted with the same function. Thus, in order for the encryption to remain secure, *both the encryption function and the decryption function have to be kept secret.*

Encryption schemes with this property are called **private key encryption**, or **symmetrical encryption**, as encryption and decryption play a symmetrical role, and both need to be kept "private" (i.e., secret).

By contrast, **public key encryption**, or **asymmetrical encryption**, is an encryption scheme where the encryption function f is such that knowing f does not give away the decryption function f^{-1}. Hence the encryption function f can safely be made public (in the form of the "public key"), and only the decryption function has to be kept secret (the "private key").

An encryption function f suitable for a public key encryption must have the following properties: (i) the function f must be easy to compute; (ii) its inverse, f^{-1}, must be hard to compute; (iii) there must exist a "backdoor" approach that allows easy computation of f^{-1} given an appropriate "private key".

The RSA encryption scheme: This encryption scheme, named after its inventors in the mid 1970s (Rivest, Shamir, and Adleman), has become the gold standard for public key encryption. It works as follows.

- Two primes, p and q, are generated, and *kept secret.* Typically, p and q will have around 100 decimal digits. Computationally, this is relatively easy to do, for example, by randomly testing numbers of the desired size for primality until a prime is found.
- The **encryption modulus**, $m = pq$, is computed, by multiplying the two primes. Computationally, this is a trivial task. The modulus m is *made public.*
- A **public encryption exponent**, e, is chosen, subject to the condition $(e, \varphi(m)) = 1$. A common choice is $2^{16} + 1 = 65,537$, the largest known Fermat prime.
- A corresponding **secret decryption exponent**, d, is computed by solving the congruence $ed \equiv 1 \bmod \varphi(m)$. This requires knowing $\varphi(m)$, or equivalently, the (secret) prime factorization of the modulus m. Given this prime factorization, a solution to the above congruence can be found quickly via the Euclidean algorithm.
- **Encryption and decryption:** For simplicity, assume the message M to be encrypted is a positive integer smaller than each of the primes p and q.[1] This assumption ensures that M is in the range $1 \le M < m$ (so that M is completely determined by its remainder modulo m) and also satisfies $(M, m) = (M, pq) = 1$. The encryption and decryption functions in the RSA scheme are then defined as follows.

[1] A general message can be converted to a sequence of such numbers by first encoding the characters as numerical values (e.g., ASCII codes) and then breaking the resulting numerical sequence into blocks of suitable size.

- **Encryption:** To encrypt M, compute $M^e \bmod m$. The result, expressed as the least positive residue modulo m, is the encrypted message E. The pair (e, m) consisting of the (public) encryption exponent and the (public) modulus is the public "encryption key". Anyone who has this key (i.e., knows the values of e and m) can encrypt a message.
- **Decryption:** To decrypt an encrypted message E, compute $E^d \bmod m$. The result, expressed as the least positive residue modulo m, is the original message M; see below for a proof. The pair (d, m) consisting of the private (i.e., secret) decryption exponent and the (public) modulus is the private "decryption key".

Computationally, both encryption and decryption are "modular exponentiations" that can be performed quickly using the repeated squaring trick. For example, with $e = 2^{16} + 1 = 65,537$ as exponent, computing $M^e \bmod m$ requires only 17 operations.[2]

Why it works:

- **Existence of decryption exponent d.** The decryption exponent d is defined as a solution to the congruence $(*)$ $ed \equiv 1 \bmod \varphi(m)$. For the system to work, we need to be guaranteed that such a a solution exists. Now, $(*)$ is a linear congruence of the form $ax \equiv 1 \bmod n$. By the general theory for such congruences, a solution x exists if and only if $(a, n) = 1$, and in this case there is a unique solution in the range $1 \le x \le n$. In the case of $(*)$, the condition for the existence of a solution d becomes $(e, \varphi(m)) = 1$, and since e was chosen to satisfy the latter condition, this shows that $(*)$ has a solution d. Moreover, the general theory guarantees that there is a solution d in the range $1 \le d \le \varphi(m)$.
- **Decryption returns the original message.** Here we show that the above decryption algorithm indeed returns the original message, i.e., that $E^d \equiv M \bmod m$.

  First, note that by the congruence $(*)$, there exists a nonnegative integer k such that $de = 1 + k\varphi(m)$. Then, since $M^{\varphi(m)} \equiv 1 \bmod m$ by Euler's Theorem,

$$E^d = (M^e)^d = M^{ed} = M^{1+k\varphi(m)} = M \cdot (M^{\varphi(m)})^k \equiv M \cdot 1^k = M,$$

What makes RSA (relatively) secure: The above argument shows that the RSA scheme works correctly, in the sense that the decryption function is well-defined (d always exists) and that it returns the original message.

The security of the RSA scheme, and its usefulness in practice, relies on the fact that encrypting a message using the public key (e, m) is computationally easy, while decrypting a message knowing only the public key is computationally hard, to the point of being infeasible if the modulus is in the 200+ digit range. Here is a rundown of the computational complexity of the key tasks involved:

- **Primality testing is easy:** Current algorithms can test integers of thousands of digits in fractions of a second. Thus, finding the large primes needed in the RSA scheme is easy.
- **Factoring is hard (as far as we know):** All known factoring algorithms are only effective up to a hundred digits or so. Factoring a 200+ digit integer composed of two large prime factors, such as those arising in the RSA scheme, cannot be done in a reasonable amount of time.
- **Modular exponentiation is easy:** Given an exponent e and a modulus m, computing $M^e \bmod m$ can be done quickly using the repeated squaring trick.

[2] Compute sucessively $M^2, M^{2^2} = (M^2)^2, M^{2^3} = (M^{2^2})^2, \ldots, M^{2^{16}}$, and multiply the last result, M^{2^16}, by M, reducing modulo m at each stage.

- **Computing modular inverses is easy:** Solving an equation $ax \equiv 1 \bmod n$ for x can be done quickly by applying the Euclidean algorithm to (a, n). Hence, determining the decryption exponent d via the congruence $(*)$ is easy, *provided the modulus in* $(*)$*,* $\varphi(m)$ *is known.*

- **Computing $\varphi(m)$, given m, is hard if the prime factorization of m is not known.** In fact, in the case when $m = pq$ is a product of two prime factors, computing $\varphi(m)$ is just as hard as factoring m. This makes it near impossible to compute the decryption exponent d via the congruence $(*)$ without knowing the prime factorization of m.

- **Computing $\varphi(m)$ is easy if the prime factorization of m is known.** If the prime factorization of m is known, we can use the explicit formula $\varphi(p_1^{\alpha_1} \dots p_r^{\alpha_r}) = (p_1^{\alpha_1} - p_1^{\alpha_1 - 1}) \dots (p_1^{\alpha_r} - p_1^{\alpha_r - 1})$. In particular, if $m = pq$, then $\varphi(m) = (p-1)(q-1)$.

Introduction to Analytic Number Theory Lecture Notes, Math 531

These are notes from an introductory graduate course on Analytic Number Theory I taught in Fall 2001 and Fall 2005 at the University of Illinois.

Change log

- **[1/7/13]** Corrected some minor typos.
- **[9/1/06]** Expanded exercise sections, by adding problems from recent comprehensive exams.
- **[12/7/05]** Minor corrections in all chapters. Exercises added.

Lecture Notes

- Complete Notes (197 pages, 800k)
- Table of Contents
- Chapter 0: Primes and the Fundamental Theorem of Arithmetic
- Chapter 1: Arithmetic functions I: Elementary theory
- Chapter 2: Arithmetic functions II: Asymptotic estimates
- Chapter 3: Distribution of primes I: Elementary results
- Chapter 4: Arithmetic functions III: Dirichlet series and Euler products
- Chapter 5: Distribution of primes II: Proof of the Prime Number Theorem
- Chapter 6: Primes in arithmetic progressions: Dirichlet's Theorem

---

Last modified Mon 07 Jan 2013 04:43:52 PM CST A.J. Hildebrand

MATH 213 F1: Basic Discrete Mathematics
Fall 2010
Professor A.J. Hildebrand
https://faculty.math.illinois.edu/~hildebr/213/

Final Exam Results and Course Grades

Final exam scores and course grades are now available. The highest score on the final was 200 and the median 131. 20 percent had a score of 179 or higher, and 40 percent had a score of 136 or above. Final Exam Solutions are available under this link.

You can access your scores as usual under this link. The grade shown at the end of the online score display (including plusses and minuses) is the grade that has been reported as your official course grade; it is based strictly on the total number of accumulated points and is nonnegotiable. Cutoffs between letter grades were chosen to minimize hardships and close calls. As a result, nobody was within a few points of the next higher grade, and most gaps were substantially larger.

Enjoy the holidays and have a happy and relaxing break!

Course Policies, Exams, Grades

- Course Information Sheet. The sheet handed out at the beginning of the semester. Syllabus, grading policies, office hours, etc.
- Online Scores. Log in with your Net-ID and password. The display shows the scores on all assignments and exams given out so far, and the total accumulated score. Any score discrepancies must be reported within one week of the assignment/test.
  Exam 3 scores and prefinal grades are in! Check here for more information about the grading and the score display. Solutions are available under this link.
- **Final Exam, Friday, 12/17, 1:30 pm - 4:30 pm, Room 161 Noyes.**
  - Final Exam Syllabus and Information. Click here for a syllabus and other information about the Final. The post-exam3 part of the syllabus will be finalized over the next couple of days.
- **Midterm Exam 1, Friday, 9/17.**
  - Exam 1 Results and Grading Information.
  - Exam 1 Solutions.
  - Exam 1 Syllabus.

- **Midterm Exam 2, Friday, 10/15.**
  - Exam 2 Results and Grading Information.
  - Exam 2 Solutions.
  - Exam 2 Syllabus.
- **Midterm Exam 3, Monday, 11/15.**
  - Exam 3 Results and Grading Information.
  - Exam 3 Solutions.
  - Exam 3 Syllabus.

Announcements

- **Open House Hours:** Hours are Tuesdays/Wednesdays/Thursdays, 5 pm - 6 pm (and longer if needed), 159 Altgeld Hall, beginning September 1.
- **Final Exam Date:** The officially assigned Final Exam slot for this class is **Friday, December 17, 2010, 1:30 pm - 4:30 pm.** Please keep this date in mind when making travel plans. University regulations are very strict in requiring students to take the Final at the assigned slot; in particular, you cannot take the Final at an earlier date to accommodate travel plans.

Class Diary

- **Wednesday, 12/8:** More big oh proofs.
- **Monday, 12/6:** Section 3.2. Growth of functions. Formal definition of "big oh" notation. Examples of big oh proofs.
  Read: Section 3.2, through p. 186. Do Problems 2(a)(c)(e), 4, 14(a)(c)(e) in 3.2. (Solutions to these problems will be distributed on Wednesday.)
- **Friday, 12/3:** Euler's formula for the number of regions in a planar graph. Degree of a region. Proof of nonplanarity of the graphs K_5 and K_3,3.
- **Wednesday, 12/1:** More on Euler and Hamilton circuits. The n-cube graph Q_n; Gray codes. Started 9.7. Definition of planar graphs. Examples. The "3 houses and 3 utilities" problem.
  Read: Section 9.7.
- **Monday, 11/29:** Euler and Hamilton circuits and paths. The Konigsberg Bridge Problem.
  Read: Section 9.5.
- **Friday, 11/19:** Connectivity in graphs. Paths and circuits. Length of a path. Connected component of a graph. Euler circuits. Bacon numbers and Erdos numbers.
  Read: Take advantage of the break to read up on graph theory (Chapter 9), especially if you missed Friday's class. The material covered on Friday corresponds to Sections 9.4 (Connectivity) and 9.5 (Euler/Hamilton circuits). After the break, I plan to spend a bit more time on 9.5, then move on to 9.7 (Planar Graphs).
- **Wednesday, 11/17:** Special types of graphs (complete graph, cycles, bipartite graphs, etc.). Graph isomorphisms.
  Read: Section 9.3.

Graded HW Assignment 10.

- **Monday, 11/15:** Midterm Exam 3.
- **Friday, 11/12:** Started Chapter 9. Formal definition of a graph and some examples. Directed graph, multigraph, and directed multigraph. Vertices and edges. Adjacent vertices. Degree of a vertex. The Handshake Theorem.
  Read: Sections 9.1 and 9.2.
- **Wednesday, 11/10:** More on equivalence relations. Equivalence classes. Connection between equivalence relations and partitions. Congruence modulo m.
  Read: Section 8.5
- **Monday, 11/8:** More examples of relations. Equivalence relations (8.5). Graph of an equivalence relation.
  Graded HW Assignment 9.
- **Friday, 11/5:** Started Chapter 8, Relations. Definition of a relation on a set A, and some examples. Representation of relations as digraphs, and as 0-1 matrices. Reflexive, symmetric, antisymmetric, and transitive properties.
  Read: Sections 8.1 (Examples 1 - 17) and 8.3 (Examples 1 - 3, 7 - 10). (Skip the final part of 8.1, "Combining relations", and Examples 4 - 5 in 8.3, which depend on this part.)
- **Wednesday, 11/3:** Wrapped up Chapter 7. Application of inclusion/exclusion to divisibility questions and the sieve of Eratosthenes.
- **Monday, 11/1:** More applications of inclusion/exclusion. Permutations with given number of fixed points. Number of onto functions. Word/string counting problems.
  Read: Sections 7.5 and 7.6. In Section 7.6 you can skip Example 1 (which gives an application of I/E to donut-counting problems), but you need to be familiar with the other applications in this section: derangements, onto functions, sieve of Eratosthenes. All of these have been (or will be) covered in class, and in the homework problems.
  Graded HW Assignment 8. **Note:** Section 7.8 refers to the "Supplementary Exercises" section in Chapter 7.
- **Friday, 10/29:** The Inclusion/Exclusion Principle, general case. Statement of the I/E formula for n sets, and first applications. Derangements.
- **Wednesday, 10/27:** Finished Section 7.4 and started Section 7.5/7.6, on the Inclusion/Exclusion Principle. Inclusion/Exclusion for 2 and 3 sets.
- **Monday, 10/25:** Wrapped up Section 7.2 and started 7.4. Solutions of nonhomogeneous recurrences. Solving recurrences by generating functions.
  Read: Section 7.4, p. 493 - 495. This part of 7.4 deals with applications of generating functions to recurrences, the only application we will cover in class; you need not know the other applications covered in the rest of 7.4..
  Graded HW Assignment 7 **Note:** Problem 8 should be #31(a)(b)(c)(d) in 7.4; in the original version this was misprinted.
- **Friday, 10/22:** Section 7.2, continued. Why the characteristic equation method works. The case of repeated roots in the characteristic equation.
  Read: Section 7.2. Examples 1 - 6, 9, 10, 11, 13. (For recurrences with repeated roots you only need to know the simplest cases given in Theorem 2 and illustrated by Example 5; you can skip Theorem 4. For nonhomogeneous recurrences, you need to be able to handle the simplest cases

such as polynomials (as in Ex. 10) and pure powers (as in Ex. 11). no need to memorize the most general case given in Theorem 6.

- **Wednesday, 10/20:** Section 7.2. Classification of recurrence relations: Linear/nonlinear, homogeneous/nonhomogeneous, constant coefficients, degreee of recurrence, General theory for solving linear homogeneous recurrences via the characteristic equation method. Derivation of explicit formula for Fibonacci numbers.
- **Monday, 10/18:** Solving recurrence relations, overview. Solution by iteration methods.
  Read: Section 7.1.
  Graded HW Assignment 6
- **Friday, 10/15:** Midterm Exam 2.
- **Wednesday, 10/13:** Started Chapter 7. Recurrence relations. Examples of deriving recurrence relations: Bitstrings with no consecutive 0's, ways to climb n stairs in single or double steps, regions in plane generated by n lines.
  Read: Section 7.1.
- **Monday, 10/11:** Independence, conditional probability, Bayes' Theorem.
  Read: Sections 6.2, 6.3. (In 6.2 you can skip the parts on Random Variables, Monte Carlo Algorithm, and the Probabilistic Method. In 6.3 you only need to know the first (simple) version of Bayes' Theorem, Theorem 1, not the general version, Theorem 2.)
  Do: Section 6.2: 7(a)(c)(e), 19, 31, 35; Section 6.3: 1, 5 (13 in 6.3 has been dropped since this requires the generalized Bayes' Theorem.)
- **Friday, 10/8:** General probability assignments, success/failure (Bernoulli) trials.
  Read: Section 6.2 (skip the parts on Random Variables, Monte Carlo Algorithm, and the Probabilistic Method).
  Do: Section 6.2: 7(a)(c)(e), 19, 31, 35;
  Section 6.3: 1, 5, 13
  (Because of the exam next Friday there will be no graded assignment that week. The above problems are practice problems you should do on your own. Solutions will be provided before the exam.)
- **Wednesday, 10/6:** Elementary probability, continued. Birthday type problems.
- **Monday, 10/4:** Started Chapter 6 (Discrete Probability). Probabilistic terminology: Sample space, events, outcomes. Definition of probability in equally likely outcome spaces. Examples: Poker probabilities, dice probabilities,
  Read: Section 6.1 through Example 8.
  Graded HW Assignment 5
- **Friday, 10/1:** Variations of the donut problem, counting solutions to equations.
  Read: Section 5.5 through p. 375 (including the section on permutations with indistinguishable objects.
  Do: Section 5.5: 6*, 8*, 9, 10(a)(c)*, 11, 12*, 14*, 15, 16*, 20*, 25, 26*, 30*, 32*, 46*.
  (Problems marked by an asterisk will be part of the next graded hw assignment. Most of other problems are "twins" to adjacent asterisk problems, i.e., problems of the same type, with answers or solutions in the back of the book.)
- **Wednesday, 9/29:** Combinations with repetitions. The donut problem.
  Read: Section 5.5 through p. 375.

- **Monday, 9/27:** Binomial coefficients and the Binomial Theorem. More combinatorial problems.
  Read: Section 5.3/5.4.
  Graded HW Assignment 4
- **Friday, 9/24:** Combinatorial problems, continued..
- **Wednesday, 9/22:** More pigeonhole examples.
  Read: Section 5.2. Study Examples 1-8 if you have not done so.
- **Monday, 9/20:** Sections 5.1. Basics of counting. Word counting problems. Complement trick, inclusion-exclusion.
  Section 5.2. The Pigeonhole Principle.
  Read: Section 5.1, Examples 1-8, 10, 11, 12, 15-18. Section 5.2, Examples 1-8.
  Do: See the graded hw below. For additional practice, do some of the odd-numbered problems in 5.1 (except those involving divisibility since we haven't covered that yet).
  Graded HW Assignment 3 (due Friday 9/24).
- **Friday, 9/17:** Midterm Exam 1. I expect to get the exam graded this weekend. and have scores online by Sunday.
  UPDATE (9/18, 5:45 pm): Scores are up! Click on the links above to access your scores and for exam solutions and grading information.

  Meanwhile, here is an assignment for Monday:
  Read: Section 5.1. In particular, study Examples 1-8, 10, 11, 12, 15-18. Most of these examples (and the hw problems below) are on the easy side, but they form the foundations for the more difficult counting problems that come up in later sections in this chapter. In order to be prepared for the more difficult problems it is essential that you first gain experience with these easier questions through plenty of practice.
  Do: 5.1: 8*, 14*, 16*, 17*, 23*, 24*, 28*, 30*, 34*, 38*, 39*, 43*. (Problems marked by an asterisk will be part of Graded HW 3 (due Friday, 9/24).
- **Wednesday, 9/15:** Wrapped up the discussion on induction. Examples of flawed induction proofs illustrating some common errors and pitfalls. Review of general structure of an induction proof, and the logic behind it.
  Do: No additional assignments.
- **Monday, 9/13:** More induction proof examples.
  Do: No additional problems. Finish the assignments from last time, from the induction handouts.
  Handout: Sample Induction Proofs.Model solutions to problems on the two induction handouts.
- **Friday, 9/10:** More induction proofs. Strong induction. Fibonacci numbers.
  Read: Section 4.1. In particular, read the introductory section, and study Examples 1-10.
  Do: Problems on the induction handouts. From Rosen, do 7, 13, 15, 19, 21, 23, 25 in Section 4.1 and 13, 15 in Section 4.3.
  Handout: Worksheet: Induction Proofs, II
- **Notes on HW 2:** You can ignore the last problem from 2.4, (\#8, Problem 34(c)(d)), which turns out to be harder than it originally seemed (it requires a method similar to that proving countability of rations). (The preceding problem (\#7, Problem 32(a)(b) in 2.4), by contrast, is very easy.)
- **Wednesday, 9/8:** Started Sections 4.1/4.2, on Induction. The Induction Principle, and why it

works. The structure of an induction proof, base step, induction step. Examples (to be continued Friday and Monday).
Read: Section 4.1. In particular, read the introductory section, and study Examples 1-10.
Do: 4.1: 7, 13, 15, 19, 21, 25.
Handout: Worksheet: Induction Proofs

- **Friday, 9/3:** Cardinality, countable sets. Two famous results with cool proofs: the countability of the rations, and the non-countability of the reals. Fun application of these ideas: Hotel Infinity.
Graded HW Assignment 2 (due Friday 9/10).
Monday is Labor Day, so the next class will be Wednesday. If you want a fun and completely non-technical read for the long weekend, here is a much embellished short story version of the "Hotel Infinity" problem. For a much drier analysis of this problem, read the the Wikipedia article on the subject.
Read: Finish up 2.4. The final part of this section covers cardinality and countability.
- **Wednesday, 9/1:** Section 2.4. Formal definition of a sequence. Arithmetic and geometric progressions. Summations. Formula for sum of finite geometric series, and sum of first n positive integers (memorize these!). Summation and product notation.
On-Line Encyclopedia of Integer Sequences. A searchable database of 100,000 sequences, and arguably the most amazing math resource on the web. It comes with its own WebCam. Check out the puzzle sequences.
Read. Section 2.4 through p. 158.
Do (non-graded). Section 2.4: 1, 3, 13, 15, 17, 27.
Graded HW 2, preview: From Section 2.4, the assignment will include Problems 4(a)(d). 14(a)(d), 16(a)(d), 18(a), 28, 30, 32. I will add a few more problems. The full assignment will be given out Friday, 9/3. instructions.
- **Monday, 8/30:** Section 2.3, continued. The graph of a function as a subset of A x B. Surjective, injective, and bijective functions. Compositions and inverse functions. Bijections from Z to subsets of Z.
Graded HW Assignment 1 (due in class Friday, 9/3)
- **Friday, 8/27:** Section 2.3 (Functions). Formal definition of a function. Representation by arrow diagram. "Black box" interpretation. Domain, codomain, image.
Read. Section 2.3.
Do (non-graded): Section 2.3: 1, 13, 15, 17, 19.
The first graded assignment will be handed out Monday and due Friday, 9/3. See the preview in Wednesday's class.
- **Wednesday, 8/25:** Section 2.2 (Set Operations): intersection, union, difference, complement. Venn diagrams. Set identities (see table p. 124). Formal proofs of set identities (as in Example 10). Membership tables (as in Example 13).
Read. Section 2.2 (you can skip the final part, on computer representations of sets).
Do (non-graded hw). Section 2.2: 1, 3, 13 (Proof*), 17 (Proof*), 25, 27, 29. *For these proof problems, give both a formal proof as in Example 10 and a proof via membership tables as in Example 13.
Graded HW Assignment 1, Preview: The first graded assignment will be given out Monday

and due Friday, 9/3. Here is a preliminary version of this assignment: Section 2.1: 8(a)(c)(e)(g), 28(a)(d). Section 2.2: 4, 14, 15 (Proof*), 24 (Proof*), 26. Section 2.3: 2, 12, 13, 16, 18, 32. For the proof problems a formal proof as in Example 10 and a membership table proof as in Example 13 is required. Depending on how far we get in class, I may make small adjustments in the assignment (especially on Section 2.3). The official assignment will be given out Monday, along with some instructions.

- **Monday, 8/23:** Course overview. Handed out Course Information Sheet. Section 2.1 (Sets). Sets, elements of sets, subsets, power sets, empty set. Cartesian products.
  Read. Preface and Section 2.1 through p. 118.
  Do (non-graded). Section 2.1, 1, 3, 5, 7, 9, 17, 19. (These are easy quickies. Answers can be found in the back of the book. These daily assignments are not be turned in. The first graded assignment will be given out next Monday and will be due Friday next week (9/3).)

Last modified: Sat 18 Dec 2010 03:35:03 PM CST A.J. Hildebrand

Fall 2017 Courses

Collapse All

ABE 199 CHP: Water in the Global Environment, *Prasanta Kalita*

55376 | 4:00 - 5:20 p.m. | TR | 204 Agriculture & Engineering Science Building | 3 Hours

---

"Water in the Global Environment" proposes to enhance students' understanding and appreciation of the impact water has globally, including various cultures around the world. Students will be encouraged to step outside their traditional thinking and become knowledgeable about how water availability and quality affect the day to day lives of people. Without water, or suitable water, cultural infrastructure is destined to fail. Water is arguably the most precious resource in the world, and the fact that it is non-renewable provides additional value that students will become well versed in. Water quality and its impact on global environment will be explicitly covered. Students develop in-depth analyses of case studies, which will examine the historical and current water-related issues and the solutions utilized to tackle the issues in various parts of the world (i.e., Africa, Asia, Europe, Latin America, Middle East, South America, USA). This course's goal is to not only educate students on one of the most important and critical areas of concern in the world today, but to motivate them to use enhanced knowledge to make an impact both locally and globally.

***This course has been petitioned for general education credits. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media for Physical Sciences and Non-Western.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

Instructor: Prasanta Kalita is a Professor and division leader of the Soil and Water Resources Engineering Program in Agricultural and Biological Engineering and an Assistant Dean of Research in the College of ACES. His research focuses on hydrology, watershed quality, modeling erosion and sediment control, and he was recently honored at the 2013 American Society of Agricultural and Biological Engineers (ASABE) International Meeting as an ASABE Fellow.

ARCH 199 A1: Architecture + Innovation, *Kevin Erickson*

66914 | 11:00 a.m. - 12:20 p.m. | MW | 102 A Architecture | 3 Hours

We are surrounded by design everyday - both good and bad - this course will delve into the world of architecture and design looking at how we shape the built environment and how it shapes us. Through a series of lectures, readings, workshops, fieldtrips, and design exercises students will investigate everything from small-scale, commonplace objects to buildings to the design and planning of large cities. During the semester, topics such as: design and innovation, what makes good design, and the history of our built environment, will challenge students to think about the world around us through a new creative lens.

This course has not been petitioned for general education credit.

The goal of this course is to give students a basic understanding of our physical world, its materiality, and how things are made but more importantly provide them with knowledge, tool sets, and critically thinking necessary to design and create for our future. The process of design and making is quite rewarding and students will be asked to engage in this through two design exercises during the semester. This process will involve articulating a clear concept, developing that concept through sketches, drawings, models, prototypes, and other visual representations. Technologies such as computer modeling, laser cutting, 3D printing, and robotics will used to help realize student projects.

Instructor: Kevin Erickson is an architect and founding principal of KNE studio, based in New York City, and an Associate Professor at the University of Illinois, where he is Chair of the Detail + Fabrication Program Area within the School of Architecture. The work of KNE studio is fixated on simple and forward thinking design ideas that create new value on multiple scales. In 2013, KNE studio was invited by actress Bette Midler's NYRP Foundation to design a boathouse and outdoor classroom in upper Manhattan. In 2012, they won the Warming Huts Design Competition to build a winter pavilion alongside Frank Gehry, and previously received an honorable mention in the Tokyo Fashion Museum on Omotesando Competition, and were a finalist in the urbanSHED International Design Competition. The studio has received an AIANY Design Award, an ACSA Faculty Design Award, and the Next Landmark Award at the 2012 Architecture Biennale in Venice. Recently, Kevin co-curated 5x5: Participatory Provocations, a traveling exhibition featuring work by 25 young American architects and was invited to serve on the Van Alen Institute’s Program Leadership Council.

ART 199 GD: Painting and Collage Creations
Glen Davies

40483 | 12:00 - 1:50 p.m. | TR | 330 Art & Design Building | 3 Hours

---

This course is open to all, and requires no prerequisites. In this course we will learn how to create our own visual language as we explore color, composition, and content. Through the completion of four or five projects we will investigate and explore the uses of paint and traditional art materials along with techniques that include collaged elements.

Each of the studio assignments will address issues that broaden our knowledge of painting and the materials and techniques that are used to achieve this goal. Through the observation and examination of the world we live in, and the experimental use of paint and other materials in our assigned projects, we will educate our eyes and learn to make visual associations that yield content and encourage further investigation.

This course has been petitioned for general education credits.

Additional fee is required for this course.

Instructor: Glen Davies attended the School of the Art Institute of Chicago, where he was influenced by the homegrown pop genre imagism. This helped set the stage for his recurring art themes: spiritual conflict, grotesque figural fantasies and complex psycho-dramas. After spending time traveling with circuses and carnivals, Davies worked as a billboard artist and sign painter before opening a mural painting business. After completing a BFA at Drake University and an MFA in painting from the University of Illinois, Davies has divided his time between studio pursuits and a variety of alternative employments, including circus/carnival showpainter, sideshow banner artist, professional muralist, curator, and educator. Visiting artist and lecture duties have taken Davies to numerous colleges, universities and museums.

ARTD 209 KG1: Rigidity and Flexibility in Japanese Arts/Culture, *Jennifer Gunji-Ballsrud*

54029 | 1:00 - 3:40 p.m. | W | Japan House | 3 Hours

The types of arts introduced in this class are Chado: the Way of Tea, Kado: the Way of Flower, Shodo: the Way of Calligraphy, The Sodo: the Way of Kimono, Kado: the Way of Poetry, and Jindo: the Way of Human beings. As I have listed, many of the Japanese traditional arts have do, as their suffix. Do is translated as Tao in Chinese. In Japanese, it is translated into the path and connotes that it is an infinite, unlimited path, yet it is the constant goal of spiritual yearning and striving. Thus, it should be noted that traditional Japanese arts place the emphasis on spiritual attainment more so than technical attainment, and require actual practice or direct experience to gain insight. Therefore, in this class, students are not only required to read textbooks and other materials, but also have hands-on experiences with various time-honored Japanese arts. My hope is that students will learn the importance of rigid discipline and basic principles; and, thus, eventually, they will be able to apply those principles to their own specialized fields and life. There is a $50.00 materials fee for this course.

Campus has granted every section of this course with general education for Non-Western.

Additional fee is required for this course.

Instructor: Jennifer Gunji-Ballsrud is an Associate Professor and the former program Chair of Graphic Design Program of the School of Art and Design. She participated in the graphic design program for 13 years. She was an art director and one of the originators of Ninth Letter Literary and Arts Journal and website which has been

graced with design awards from the AIGA, SPD, UCDA, TDC and from the Red Dot International Design. Her design work has been published in Print magazine, How, Step Inside Design, 365: AIGA Design Annual, Creative Quarterly and many others. She has been studying the Urasenke Way of Tea since 1990 under various teachers. She has earned the Wakindate level as an intermediate student in the Urasenke Foundation. She has been teaching university courses for Japan House for the past 7 years. For the last 5 years, she has been the full-time Director of Japan House. On March 31, 2004, she received the Commendation in Commemoration of the 150th Anniversary of the United States-Japan Relationship from Foreign Minister of Japan, Jyunko Kawaguchi. On June 3, 2004, she also received a Certificate of Thanks from Sen'ei Ikenobo, 45 th Generation Headmaster of the Ikenobo Ikebana School. Both awards recognized her contribution to promote and strengthen the ties of friendship and goodwill between the U.S. and Japan. She also received the University of Illinois Campus Award for Excellence in Undergraduate Teaching, 2003; The Broadrick-Allen Award for Excellence in Honors Teaching, 2003, an award presented annually by the Campus Honors Program; and The Senior 100 Faculty Award, presented by the University of Illinois Alumni Association.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

ART 199 GDC: Experimental Materials in Painting, *Glen Davies*

31410 | 9:30 - 11:20 a.m. | TR | 330 Art and Design Building | 3 Hours

---

In this course, we will explore the use of materials, both traditional and unconventional, as a way of increasing the scope of our visual language. Just as the manipulation of paint and the exploration of its many properties helps to reveal something about ourselves and the world, combined materials, found objects and collaged sculptural forms provide another essential arena to examine. The choices we make in combining materials forces us to examine the many aspects of content and metaphor. These key concepts will be our focus here.

***This course has been petitioned for general education credit. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media at this time for Literature and Arts.

Additional fee is required for this course.

Instructor: Glen Davies attended the School of the Art Institute of Chicago, where he was influenced by the homegrown pop genre imagism. This helped set the stage for his recurring art themes: spiritual conflict, grotesque figural fantasies and complex psycho-dramas. After spending time traveling with circuses and carnivals, Davies worked as a billboard artist and sign painter before opening a mural painting business. After completing a BFA at Drake University and an MFA in painting from the University of Illinois, Davies has divided his time between studio pursuits and a variety of alternative employments, including circus/carnival showpainter, sideshow banner artist, professional muralist, curator, and educator. Visiting artist and lecture duties

have taken Davies to numerous colleges, universities and museums.

CMN 220 CHP: Communicating Public Policy, *Grace Giorgio*

54930 | 2:00 - 3:20 p.m. | TR | 1051 Lincoln Hall | 3 Hours

Place-making is a complex and contested practice with ramifications for our social, cultural, political, and economic experiences. Drawing from the social science disciplines of history, geography, and communication, CMN 220: Communicating Public Policy: Our Cities/Ourselves introduces students to how public policy, the decisions we make and how we communicate them, affects the places in which we live. Grounded in the theories and practice of argumentation, persuasive writing, and debate, the course examines rural, urban, suburban communities as well as college towns, and how public policy local, regional, and federal has contributed to the making of such communities. The course asks students to reflect on where they live, have lived, and hope to live, while considering how their own expertise and interests can be brought to the public policy table of community development and enhancement. Thus, CMN 220 takes students out of familiar territory into new places, places they may not have considered as important or relevant to their lives, but as they will learn, are connected to their fields of study as well as to who they are and who they want to become. CMN 220 culminates in a public project of small group problem solving. Students will join together to tackle a problem facing a real community and write a public policy proposal and present it to the public via a podcast or other multimedia format of their choice. In this sense, the work done in this class extends beyond the classroom and into the public realm.

***This course satisfies the advanced comp gen ed from campus. In addition, it has been approved for additional gen ed credit for Social Sciences by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media.

Instructor: Dr. Giorgio has been teaching in the Department of Communication since she arrived on campus as a graduate student in 1995. In 2001, she began teaching fulltime for the University, developing and teaching courses in gender communication, public policy and sustainability, and the geography of culture. Dr. Giorgio began teaching for Campus Honors in the fall of 2012, launching a course on place making, Communicating Public Policy: Our Cities/Ourselves (CMN 220). In 2013, Dr. Giorgio received the Chancellor s Undergraduate Teaching Award. In the fall of 2015, she received two Provost Office grants to develop and launch Writing Fundamentals, an online, interactive grammar program for Illinois writing courses. Her research interests include an experimental use of qualitative research methods to investigate the intersection of self, culture, and the public sphere. Dr. Giorgio also directs Oral and Written Communication (CMN 111/112) and manages her department's teaching internship. She serves on undergraduate distinction projects, oversees independent studies and internships. Her film production background has helped her guide students with making public projects such as videos, podcasts, and performance installations.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

CWL 395 NB2: Memory: Between Psychoanalysis & Neuroscience, What Literature & Film Can Teach Us, *Nancy Blake*

64894 | 11:00 a.m. - 12:20 p.m. | MW | 212 Honors | 3 Hours

Amnesia is "film noir's version of the common cold" (Lee Server, in Ava Gardner: "Love is Nothing". (2006)

A return to Freud's disagreement with Jung, in particular his plea for a material based science as opposed to occultism, should convince us that there is a continuum, rather than a change of direction, between psychoanalysis and neuroscience. Since Freud insisted that psyche and soma are isomorphic, it follows that information processing can be studied fruitfully from both the perspective of psychoanalysis and of neuroscience. Neuroscience demonstrates that because of brain plasticity, anything learned brings about an anatomical change in the brain. Finally the recent work on "mirror neurons"sets up a means of documenting the relationship between first-hand experience and spectatorship. The automated decision-making processes in the brain which are not yet fully understood, constitute what Freud and Lacan call the unconscious. In 1987, Stern conjectured that what has been called the Oedipal complex depends on the ability to construct a narrative, i.e. to integrate affect with words and images.

If, as has often been noted, literature and film has, from their beginnings been fascinated by the themes of memory and amnesia, dream and reality, the capacity to conceptualize the future based on the experience of the past, then film seems an ideal medium to explore some of the questions raised by the sciences of the mind today.

***This course has been petitioned for general education credit. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media for Literature and the Arts and Western.

Instructor: Nancy Blake is a Professor of Comparative and World Literature, Cinema and Media Studies, Women and Gender Studies and an affiliate of the Unit for Criticism and Interpretive Theory and the Campus Honors Program. She is also a trained psychoanalyst who taught in France before joining the University of Illinois. She has regularly been recognized on the ICES list for courses taught for the Honors Program and for her undergraduate and graduate courses.

ECON 102 CHP: Microeconomic Principles, *Martin Perry*

63021 | 11:00 a.m. - 12:20 p.m. | MW | 106 David Kinley Hall | 3 Hours

---

This course focuses on the fundamental concepts and analysis of microeconomics, including supply and demand, the price mechanism, costs and revenues, theories of the firm, market structures, factor and resource markets, market failure and the impact of government in promoting economic efficiency. The course examines economic decision-making by individuals and firms and encourages students to apply microeconomic tools to current economic policy problems and issues such as pollution, rent controls, farm subsidies and welfare policies.

Upon completion of the course, a student will: 1) Have a strong foundation in the theory and concepts of microeconomics 2) Make connections between real world and academic economics 3) Understand the relationship between conceptual and pragmatic applications of economics to the economic behavior of the "representative" individual, product, firm, market, price, etc. 4) Learn how to apply the tools of micro theory to policy problems and issues such as global warming, pollution, health care, and government subsidies.

Campus has granted every section of this course with general education Social Science credits.

++You will not receive additional CHP credit for ECON 102 (or ECON 103), if you have had another ECON 102 section (or same for ECON 103). ++

Instructor: : Martin Perry is Department Head and Professor of Economics at the University of Illinois at Urbana-Champaign. Dr. Perry specializes in the antitrust analysis of mergers and vertical restraints. Before joining the University of Illinois in 2011, Dr. Perry served as Professor of Economics at Rutgers University in New Brunswick, New Jersey since 1989. In addition, Dr. Perry held positions in research groups at Bell Telephone Laboratories and Bell Communications Research; has visited the Wharton School of Management at the University of Pennsylvania; and is a Research Affiliate at the Institute of Economic Analysis in Barcelona, Spain. During 2004, Dr. Perry served as the Chief Economist of the Federal Communications Commission. In that position, he worked on the investigation of the merger between Cingular Wireless and AWE Wireless. Dr. Perry has published in the areas of vertical integration, resale price maintenance, exclusive dealing, monopolistic competition, oligopoly, and mergers. His current research projects include vertical foreclosure, aftermarket pricing, and applications of auction models to industrial organization. Dr. Perry has consulted for federal and state agencies on mergers in various industrial product markets and the casino gaming market in Atlantic City. Dr. Perry has also consulted on private antitrust cases involving aftermarket pricing, exclusive dealing, price discrimination, and casino gaming.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

FAA 110 D: Exploring Arts and Creativity, *J.W. Morrissette & Brad Mehrtens*

66964 | 10:30-11:50 a.m. plus outside course events | R | 323 Education | 3 Hours

---

High and street art, tradition and experimentation, the familiar and unfamiliar, international and American creativity provide this course's foundation. Students will attend performances and exhibitions, interact with artists, and examine core issues associated with the creative process in our increasingly complex global society. Faculty from the arts, sciences, humanities, and other domains will lead students through visual arts, music, dance, and theatre experiences at Krannert Center and Krannert Art Museum to spark investigation and dialogue.

The class meets twice per week: once a week for discussions, and a second time to attend performances and/or exhibitions at Krannert Center for the Performing Arts and/or Krannert Art Museum. Event dates will vary. Admission to all events will be provided without charge to students enrolled in the course.

This course satisfies the General Education Criteria in Fall 2017 for a Literature and the Arts course.

Instructor: Brad Mehrtens - Instructor and Advisor, Department of Molecular and Cellular Biology

Brad earned his bachelor's in biology from Truman State University, and his master's in microbiology from Illinois. His research interests include educational pedagogy; course design; and assessment; his advising interests include transitions for freshmen and transfer students; preparing for professional or graduate programs; understanding the undergraduate research experience; acknowledging and addressing academic or personal issues. As for hobbies, Brad enjoys acting, theatre, movies, music, and sports.

Instructor: J.W. Morrissette - Assistant Head, Department of Theatre

J.W. has served in the Department of Theatre for 21 years. He has also served as the chair of the BFA Theatre Studies Program as well as the assistant program coordinator for Inner Voices Social Issues Theatre. He earned his BFA in Acting at Otterbein College in Westerville, Ohio, and both his MFA in Acting and MA in Theatre History at the University of Illinois. J.W. has taught and directed for the past 17 years with the summer Theatre Department at Interlochen Center for the Arts, has directed and taught at Parkland College, and teaches acting, directing, and Introduction to Theatre Arts at Illinois. He has been integral in developing components for the online course offerings in the department, as well as supervising all senior Theatre Studies Thesis Projects.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

HIST 295 A: Darwin & the Darwinian Revolution, *Mark Micale*

43871 | 12:30 - 1:50 p.m. | MW | 212 Honors | 3 Hours

It is universally acknowledged today that the ideas of Charles Darwin initiated one of the most profound and provocative transformations in all of human thought, science, and culture. This is an Honors seminar about the intellectual origins, scientific content, and social, cultural, and religious impacts of Darwinian evolutionary theory in the nineteenth and twentieth centuries. Our core subject will be Darwin's life, work, and world. The course also provides a historical case study in the development and diffusion of radical scientific ideas and explores the origins of the most successful and comprehensive theory in the contemporary life sciences. We will study Darwin and several other scientists of the Victorian age, followed by an examination of their influence on such diverse cultural fields as politics, philosophy, social theory, literature, gender relations, and international affairs, as well as religion.

***This course has been petitioned for general education credit. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media for Historical and Philosophical Perspectives.

Instructor: Mark S. Micale is Professor in the Department of History, where he specializes in modern European history, cultural and intellectual history, and the history of science and medicine. After completing graduate school at Yale in 1987, he taught at the University of Manchester in Britain and then joined the U of I community in 2000. He is the author of several books dealing with the history of medicine, especially psychiatry and psychoanalysis, and modern France. Micale has won a number of departmental and campus-wide teaching prizes, and in 2012-23 served as the U of I Distinguished Teacher-Scholar. He has taught three times before in the Campus Honors Program.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

HUM 395 A: Biology and Society: From Organism to Politics, *Samantha Frost/Judith Rosine Kelz*

67677 | 11:00 a.m. - 12:20 p.m. | TR | 212 Honors | 3 Hours

This course will draw on the fields of history, science and technology studies, political theory, philosophy, biology, sociology and environmental studies to introduce students to the emerging field of bio-humanities. Its aims are twofold: to come to term with the ways that science and society shape one another and to teach students how dialogue between diverse academic fields can help us to create a broader understanding of the socio-political and ethical challenges we face today.

***This course has been petitioned for general education credit. It has been approved by ACES, AHS, BUS, EDU, FAA, LAS, and Media for Historical and Philosophical Perspectives and Social Sciences. ENG has approved this for ONLY Social Sciences credit.

This interdisciplinary course will fulfill 3 credit hours towards General Education requirements in Humanities & Arts (Historical and Philosophical Perspectives) and in Social and Behavioral Science (Social Science). The course satisfies the General

Education Historical and Philosophical Perspectives requirement in that it presents the historical continuities and discontinuities in the way that scientific findings have shaped political life and the way that the demands of politics have shaped scientific research; it facilitates an appreciation of how contemporary scientific questions and critical engagements with science have emerged from the past; it fosters cross-disciplinary dialogue as a counterweight to the narrowness of disciplinary expertise; it stimulates questions concerning the nature of the human; it engages questions of ethics and principles in scientific practice and in the instrumentalization of scientific findings in social and political life; and it fosters critical thinking and refined judgment with regard to the interaction between science and society.

In addition, the course satisfies the General Education Social Science requirement in that it traces the interactions and the effects of interactions between scientists, research institutions and practices, philosophers, political activists, and political and legal institutions; it provides historical, social, and political-economic context for scientific research; it deploys specific instances as well as historical patterns to explore how scientific research findings and biotechnological innovations provoke and are bound up with transformations in philosophical, ethical, and political understandings of social and political life; and it provides examples and opportunities to undertake interdisciplinary research in this field.

Through this course, students will also learn different modes of critical and interpretive analysis, historical contextualization, network analysis, discourse analysis, and develop a working familiarity with research fundamentals in molecular and cellular biology.

Instructor: Dr. Rosine Kelz has taught undergraduate and graduate courses in a variety of academic settings and consistently received excellent teaching evaluations. Most recently, she taught Social Theory in the M.A. program 'Critical Theory and the Arts' at the School of Visual Arts, New York, and undergraduate courses in political and social theory at the University of Hamburg and Humboldt University, Germany. She is accredited as an Associate Fellow of the Higher Education Academy by the Developing Learning and Teaching Program of the University of Oxford.

KIN 340 SP1: Sociology & Psychology of Physical Activity, *Steve Petruzzello*

64961 | 1:00 - 2:20 p.m. | TR | 130 Freer | 3 Hours

Social and Psychological Aspects of Physical Activity is designed to acquaint you with how psychological and social processes and constraints influence human action in physical activity environments. The course will utilize both lecture and laboratory/ discussion formats, with ample opportunity for interaction and discussion between professor and students and among yourselves. There may be occasional guest lectures. You, as the student, should feel free (and are strongly encouraged) to ask questions, take alternate viewpoints, present supportive arguments for statements, and generally make yourself a presence in the class. This cannot be emphasized enough. Keeping your insights and ideas to yourself will deprive us all of potentially

illuminating, interesting, and useful information.

I believe in the following statement by Socrates: "I cannot teach anybody anything; I can only make them think." From you I expect: (a) commitment to excellence, that is, I don t want you to overlook other important aspects of your life, but I do expect you to do work, spend the time, and do the reading and writing (and thus, thinking) necessary to be successful in this course; (b) self-motivation; and (c) initiative and critical thought. If you leave my classroom and have acquired a stronger ability to think, I will have done my job.

This course satisfies the advanced comp gen ed from campus.

Instructor: Steven Petruzzello is an Associate Professor in the Department of Kinesiology and Community Health. He received his Ph.D. in Exercise Science, Psychology of Exercise and Sport from Arizona State University in 1991. He began his career at UIUC in 1991 and has served as the Associate Head for Graduate Studies, Department of Kinesiology & Community Health since 2011. He has also been a Research Scientist for the Illinois Fire Service Institute since 2005. Professor Petruzzello's research focuses on determining the mechanisms underlying the effectiveness of exercise in improving affect/emotion. The second line of research examines the physiological and psychological aspects of firefighting. Professor Petruzzello has been awarded the College of Applied Health Sciences Undergraduate Teaching Faculty Award, the Campus Award for Excellence in Undergraduate Teaching, and has consistently been named to the List of Teachers Ranked As Excellent By Their Students.

LING 240 B: Language in Human History, *Hans Heinrich Hock*

40360 | 12:30 - 1:50 p.m. | TR | 212 Honors | 3 Hours

Whose past is it? The uses and misuses of linguistics, prehistory, genetics/genomics, and archaeology in national and group self-definition and in the exclusion of others. We will focus especially on the Aryan issue in four contexts: Nazi ideology and the "Aryan Myth", neo-fascist movements in Russia, current ideological movements in South Asia, and ideological interpretations of the Tarim Mummies of Xinjiang. In the course of evaluating these issues, we will discuss the question of scientific methodology and the responsibility of historians, linguists, geneticists, and archaeologists to address misuses or deliberate misinterpretations of their results by nationalist and racist ideological movements.

This course satisfies the Hist & Philosoph Perspect gen ed from campus.

Instructor: Instructor: Hans Henrich Hock is Professor Emeritus of Linguistics and Sanskrit, University of Illinois. He has taught and done work in Indo-European comparative and historical linguistics with major focus on Sanskrit, as well as on language and ideology. Major publications include Principles in historical linguistics (1986, 21991), Language history, language change, and language relationship (co-

author with Brian D. Joseph, 1996, 2nd ed. 2009, 3rd e. In Preparation), Studies in Sanskrit syntax (ed., 1991), An early Upanishadic reader (2007), The languages and linguistics of South Asia (ed. with Elena Bashir, 2016). Honors include recognition as Vidyasagara by Mandakini, 10th World Sanskrit Conference, Bangalore (1997), election as Fellow of the Linguistic Society of America (Class of 2013), and the Sukumar Sen Memorial Gold Medal for work in historical and comparative linguistics (Asiatic Society, Kolkata, India, 2015).

+ Section is now full; please contact CHP staff to be added to a waitlist.+

MATH 199 A: Probability and the Real World, *A. J. Hildebrand*

47745 | 4:00 - 5:20 p.m. | TR | 243 Altgeld | 3 Hours

Among all mathematical fields, probability and statistics are perhaps the most relevant in the real world and the most frequently encountered in daily life, but also the most common source of misconceptions and fallacies by the general public. In this course we will explore this subject using real-world examples taken from sports, society and everyday life, and selected to match the interests and background of the audience. Past editions of the course have touched on topics ranging from predictions and rankings in sports to the physics of coin tosses, the psychology of probability, fraud detection using probability; freakonomics type questions, as well as classic puzzles and paradoxes in probability that are both fun and educational.

This course has no formal prerequisites and can accommodate a broad range of backgrounds; in particular, no prior knowledge of probability or statistics is assumed. The course has very little overlap with traditional courses in probability or statistics and can serve to complement such courses. Grading will be based on participation, assignments, individual and group projects, and student presentations.

*** This class was petitioned for Gen Ed credit. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media for Quantitative Reasoning I.

Instructor: A.J. Hildebrand is Professor Emeritus of Mathematics at the University of Illinois. His research interests are in the areas of number theory, probability, combinatorics, and analysis. In his more than three decades at Illinois, Dr. Hildebrand has taught at all levels on subjects ranging from calculus to probability, actuarial statistics, number theory, and mathematical writing. In 2011 he received the Campus Award for Excellence in Undergraduate Teaching. Outside the classroom, he is involved in running the math contest program at Illinois, mentoring students in undergraduate research, and directing a summer undergraduate research program in mathematics.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

MATH 199 SFH: Spaceflight, *Julian Palmore*

66716 | 9:30 - 10:50 a.m. | TR | 212 Honors | 3 Hours

The course will explore the current state of human spaceflight, starting from the early days of Tsiolkovsky and Goddard to the later years of the American Rocket Society and the German VFR prior to World War II to the Mercury, Gemini and Apollo programs during the 1960s and 1970s and the Space Shuttle - International Space Station developments since 1980. We will study the mathematics, physics, astronomy, chemistry and physiology of human spaceflight.

This course has not been petitioned for general education credit this term.

Instructor: Julian Palmore is Professor of Mathematics at Illinois and teaches courses in differential equations and probability. He studied physics at Cornell University and after graduating and commissioning he was assigned to the director's office of Wernher von Braun at NASA's Marshall Space Flight Center in Huntsville, Alabama. His first published paper was "Lunar Impact Probe" in the American Rocket Society Journal in 1961. At NASA, he worked with Ernst Stuhlinger on systems analysis of ion rockets and participated in the Apollo program and later as a test engineer on the first stage the Saturn V launch vehicle. He left NASA in 1964 to attend graduate school at Princeton University in aeronautical engineering. He studied astronomy at Yale University, specializing in celestial mechanics, and returned to Princeton as a visiting fellow. He studied mathematics at the University of California Berkeley. In his career he has solved problems of rocket flight, celestial mechanics, and spaceflight.

REL 291 A/AAS 291 A: Hinduism in the U.S., *Raj Pandharipande*

68283 / 68316 | 2:00 p.m. - 3:20 p.m. | TR | G36 FLB | 3 Hours

The course is designed to introduce students to the historical, religious, and socio-cultural dimensions of Hinduism in the United States. In particular, we will examine the following questions: a) why Hinduism was brought to the US (motivations behind the migration of Hinduism to the US), b) how and why the religion has changed in the US (transformation of Hinduism in the US), c) what role(s) Hinduism plays in the "new homeland" (for the Hindu and non-Hindu population in the US), d) what strategies are adopted for the maintenance and transmission of Hinduism in the context of the socio-cultural milieu of the US, e) how Hinduism has impacted the majority culture in the US and India, and f) how the changes in the system are authenticated (the question of Authority and the mechanism of authentication of Hinduism) in the US. Embedded in the above themes will be discussions on the issues related to the family, Hindu and/or Indian identity, intergenerational tensions, the relationship of Hinduism with other religions in the US, and the global issues of religious diversity and interethnic dialogues. The course material will include selected texts, films, and videos. Students will be required to participate in group projects, and field trips. Suggested pre-requisites: Asian mythology (Religious studies 104), Introduction to Hinduism

(Religious Studies 286) or equivalent. Familiarity of subject matter is preferred but not required; contact the instructor if there are any questions.

***This course has been petitioned for general education credits. It has been approved by ACES, AHS, BUS, EDU, ENG, FAA, LAS, and Media at this time for Historical and Philosophical Perspectives and Non-Western.

Instructor: Rajeshwari V. Pandharipande is Professor of Linguistics, Religious Studies, Sanskrit and Comparative Literature, Campus Honors Program at the University of Illinois at Urbana-Champaign. Professor Pandharipande holds two Ph.D. degrees--one in Sanskrit Literature, and the other in Linguistics. The primary focus of her research and teaching has been South Asian languages, Asian Mythology, Sociolinguistics, Sociolinguistic Methodology, Language of Religion, and Hinduism in India and in Diaspora. The three major languages of her research and teaching are Hindi, Marathi, and Sanskrit. She has coordinated the Hindi Program at UIUC (from 1986-2002) and taught Hindi and Hindi Literature at UUC and University of Chicago. She has published a textbook, Intermediate Hindi, Volumes I and II. (Co-authored with Y. Kachru) 1982, 1988. Motilal Banarsidass Publications, and her new text manuscript, Advanced Hindi (co-authored with Rajesh Kumar and Mithilesh Mishra) which is funded by the ACDS is in its final stages of completion. Professor Pandharipande has published a book of Hindi poetry, Never is a Long Time aur anya Kavitayen (A collection of Hindi poems) 1987. Banhatti Prakashan, Nagpur, India. Professor Pandharipande has published scholarly articles (over 70 in various scholarly journals, books, and encyclopedias and has delivered over 150 talks at the scholarly meetings. She was invited to teach Hindi Literature at the University of Chicago in 1999, and 2001. Additionally, she has published extensively on Marathi: (a) Marathi: A grammar of the Marathi Language. Routledge, London. 1997, (b) Sociolinguistic Dimensions of Marathi: Multilingualiam in Central India, Lincom, Germany, 2003. She has guided research at M.A. and Ph.D. levels in Sociolinguistics, Sanskrit, and Hindi at UIUC. Her research on South Asian languages is embedded in her research on larger cultural/religious context of South Asia. She has published a book, The Eternal Self and the Cycle of Samsara: Introduction to Asian Mythology and Religion. 1990 (4 editions). Ginn Press, Massachusetts and, she is currently preparing the final version of the manuscript, Language of Religion in South Asia: theory and practice (accepted for publication by Macmillan-Palgrave (London). Professor Pandharipande received the title "University Scholar" by the Chancellor for her outstanding research at the University of Illinois, and Harriet and Charles Luckman. All Campus Distinguished Undergraduate Teaching Award, and William Prokasy Award for the outstanding excellence in undergraduate teaching at the College of Liberal Arts and Sciences at the University of Illinois at Urbana-Champaign. In 2008, Professor Pandharipande received the tile, "Distinguished University Scholar/Teacher for her outstanding achievement in research and Teaching. Professor Pandharipande's teaching at the CHP has been highly appreciated by her students.

SOCW 199 A Society and the Non-Profit Sector, *Wynne Korr*

54085 | 3:00-5:50 p.m. | T | SOCW 2023 | 3 Hours

Do you want to change the world? Perhaps you are passionate about addressing homelessness, sexual violence, or environmental problems. Do you volunteer? Make donations to charities? Want to create new solutions to social problems? Much of this happens in the nonprofit sector, yet most people know little about it. Many important resources that help people in need and that enrich our lives through the arts are provided by nonprofit organizations. The nonprofit sector, also called the non-governmental or voluntary sector, meets needs not met by business or government, has demonstrated social impact, and employs large numbers of people. For example, according to the Urban Institute, in 2013 1.4 million nonprofits were registered with the IRS and contributed $905.9 billion to the US economy or 5.4% of the Gross Domestic Product.

The purpose of this course is to enable students to understand how the nonprofit sector functions and to challenge them to think about this sector critically, understanding ethical principles and values that guide non-profit work, and research evidence about effectiveness and impact.

The course may also help inform students' thinking about career choices and/or volunteer activities.

This course has not been petitioned for general education credit this term.

Instructor: Wynne Korr served for 15 years as dean of the School of Social Work. During a long career, she has provided consultation and training to nonprofits on program evaluation and impact. She developed a graduate seminar on Leadership and Social Change with a focus on the nonprofit sector and social entrepreneurship. Her current research examines how to educate students for leadership and management in the nonprofit sector, nonprofit finance, and collaborative innovations. She has had a developmental role in campus initiatives in community engagement, social innovation, entrepreneurship, and design.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

CHP Seminar 395 B: America & the War in Iraq: No Man's Land 2.0, *Peter Fritzsche*

31625 | 3:30 - 4:50 p.m. | TR | 212 Honors | 3 Hours

Campus Honors Program Seminar course (First year students are restricted from enrolling in CHP seminars)

This course will examine the origins and conduct of one of America's longest wars, the military engagement in Iraq, from 11 September 2011, to the invasion and occupation of Iraq in 2003/04, to the beginnings of the insurgency in 2004, and to the surge in 2006 which preceded withdrawal in the years that followed. It will examine the intellectual and political framework by which the failures of the American occupation can be understood as well as the historical, geopolitical, and religious origins of both

the Sunni and Shia insurgencies. A second theme will be the representation of the war: how it was conceptualized in the Bush administration, how it was reported in the media, and how it was communicated by the troops themselves. Finally, the course will zero in on the warriors themselves the small-group relations in the unit, the difficulty of identifying dispersed enemies, the dynamics of abuse and atrocity, the return home and redeployment, and post-traumatic stress syndrome. Small oral presentations to the class and small paper assignments will precede a medium-sized research project due at the end of the semester.

This course has not been petitioned for general education credit for this term.

+ Section is now full; please contact CHP staff to be added to a waitlist.+

Instructor: Peter Fritzsche has taught History at the University of Illinois for nearly thirty years. He has received Guggenheim, Humboldt, and NEH fellowships, has written seven books in German and European history including Life and Death in the Third Reich, Germans into Nazis, Reading Berlin 1900, Nietzsche and the Death of God, and Stranded in the Present, and, most recently, An Iron Wind: Europe Under Hitler. Fritzsche has served as chair of the Department of History and has been recognized for his excellence in teaching, including regular inclusion on the "Incomplete List of Excellent Teachers." He taught several courses for CHP in the late 1980s and early 1990s, including an ACDIS-sponsored course with Jeremiah Sullivan on "the United States as a Superpower" and a CHP course on the "Temptations of Fascism." In more recent years, he has taught honors courses on the Holocaust (fall 2007 and fall 2010) and on World War I (spring 2013, spring 2014, fall 2015) as well as the wars in Iraq (fall 2016). His pedagogy emphasizes the close analysis of key texts through discussion and debate and the creation of defensible interpretations of human behavior through writing and rewriting and an empatheitc understanding of narrative, documentary, and argumentative strategies. His ultimate aim is give students confidence in speaking about the world and ultimately in judging it.

CHP 395 C: Ecological Criticism, *James Treat*

55838 | 9:30 - 10:50 a.m. | MW | 212 Honors | 3 Hours

Campus Honors Program Seminar course (First year students are restricted from enrolling in CHP seminars)

The scientific, political, and economic policy debates about global environmental crisis have tended to ignore its religious, historical, and literary dimensions. This interdisciplinary seminar in the environmental humanities redresses that omission by focusing attention on the emerging fields of religion and ecology, environmental history, and literary ecocriticism. Assigned readings are drawn from representative texts covering important developments in the relevant disciplines. Research projects allow each participant to supplement our collective effort by exploring an individual interest in greater detail. Students have the opportunity to gain a basic understanding of ecological criticism; to conduct focused research on a related topic, theme, or issue; and to develop critical skills for use in educational, professional, and personal settings.

This course has not been petitioned for general education credit for this term.

Instructor: James Treat teaches courses on indigenous religious and ecological traditions and on the place of nature in contemporary criticism. His research focuses on American Indian ways of knowing, especially in the wake of imperial modernity. One of his recent projects is "Mvskoke Country", an award-winning monthly column on tribal traditions published in the Muscogee Nation News and archived at https://mvskokecountry.wordpress.com. Earlier in his career, Treat studied the native encounter with Christianity in the contemporary period. He is the author of Around the Sacred Fire: Native Religious Activism in the Red Power Era and the editor of three volumes of native literature on related themes. More information is available at https://jamestreat.wordpress.com.

*****Courses petitioned for General Education credit by Campus Honors Program will appear on student records around the middle of term. This will not appear at the time of registration. *****

+++ Please note that campus policy restricts students from using more than 12 credits of "199" courses towards your credits for graduation. CHP allows the use of all of this type of courses towards our program requirements. You are not restricted from taking as many "199" courses as you would like just beware of this graduation credit limit.+++

(217) 244-0922
chp@illinois.edu

Back to top

Campus Honors Program
1205 W Oregon St
Urbana, IL 61801

© 2015 University of Illinois Board of Trustees